Lecture: 36

LEARNING &

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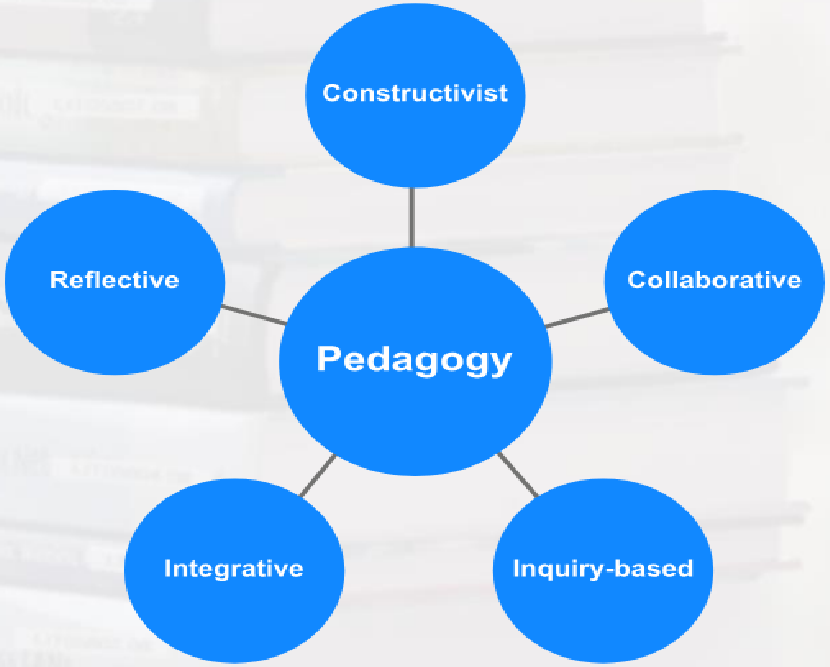
- What is Pedagogy?
- Pedagogy can be defined as a method or way of teaching any theoretical topic or academic subject.
- It has been derived from the Greek word “pedagogue” which stands for “the art of teaching children”.
- Pedagogy helps a teacher understand how learning should be facilitated.
- Pedagogy plays an important role in designing effective learning approaches and methods.
- Benefits for Learning-
- It helps improve the quality of teaching and learning.
- It assists both students and teachers in gaining an in-depth understanding of fundamental material.
- Effective teaching can help students achieve deeper learning.
- Pedagogy also enhances the student-teacher relationship.

- If teaching is the act of encouraging learning activities through discovery and acquired knowledge, pedagogy is the method of teaching, both as an academic subject or theoretical concept.
- Pedagogy enables teachers to influence student learning with an intention to capture the attention of students.
- Pedagogical methods of teaching theoretical concepts helps students to enhance their recall capacity by recollecting all the ideas that were taught to them even after a long gap.
- This is a part of innovative learning when students are able to hold on to all the ideas that were long discussed in their classrooms.
- Good pedagogy is about eliciting responses that demonstrate understanding.
- It helps the learning facilitator become agile in their approach, to match the learning processes of their students.
- Engagement in learning is increased when students or delegates have an input to the way the learning is encountered and expressed.

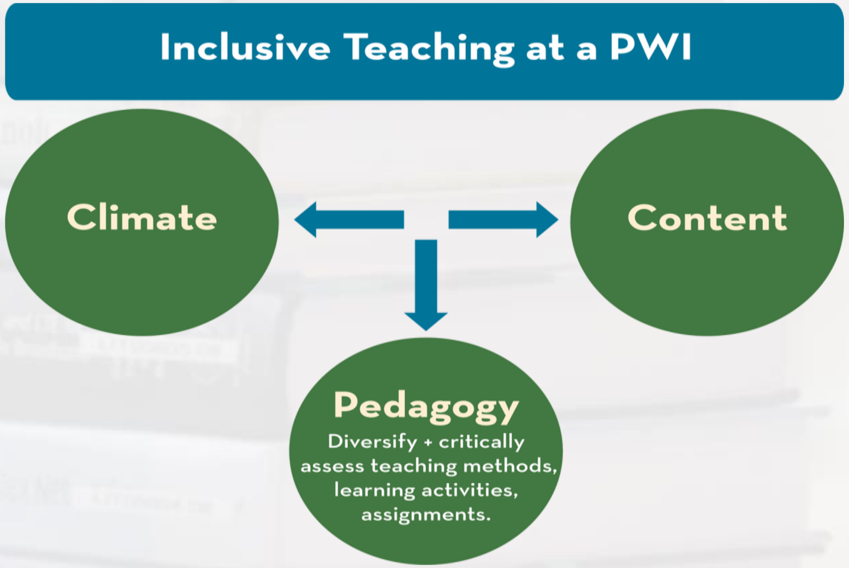
- Increased access to more metacognitive learning strategies tends to correlate to longer-term embedding of ideas and concepts,
- Delivering learning programs in ways that learners learn best makes students more likely to be engaged and have greater propensities to retain more in the future.
- Good, modern-day pedagogy involves the interconnection of learning ideas, ways and methods of training or coaching them, and the achievement of results obtained following action after the learning.
- The Five 5 Pedagogical Approaches in Teaching are:
- 1. Constructivism or the Constructivist Approach
- 2. Collaborative Approach
- 3. Inquiry-Based Approach
- 4. Integrative Approach
- 5. Reflective Approach



Types of Pedagogical Approaches

- *Constructivist Approach*
- *Reflective Approach*
- *Collaborative Approach*
- *Integrative Approach*
- *Inquiry-Based Approach*

- Constructivism or Constructivist Approach
- Constructivist teaching is based on constructivist learning theory.
- It based on the belief that learning occurs as learners are actively involved in a process of meaning and knowledge construction as opposed to passively receiving information.
- Learners are the makers of meaning and knowledge.
- Collaborative Approach
- Collaborative learning is a situation in which two or more people learn or attempt to learn something together.
- Unlike individual learning, people engaged in collaborative learning capitalize on one another's resources and skills (asking one another for information , evaluating one another's ideas, monitoring one another's work, etc.).
- More specifically, collaborative learning is based on the model that knowledge can be created within a population where members actively interact by sharing experiences and take on asymmetry roles.



- Inquiry-Based Approach
- Inquiry-based learning is a form of active learning that starts by posing questions, problems or scenarios rather than simply presenting established facts or portraying a smooth path to knowledge.
- The process is often assisted by a facilitator. Inquirers will identify and research issues and questions to develop their knowledge or solutions.
- Inquiry-based learning includes problem-based learning, and is generally used in small scale investigations and projects, as well as research.
- The inquiry-based instruction is principally very closely related to the development and practice of thinking skills.
- Integrative Approach
- Integrative learning is a learning theory describing a movement toward integrated lessons helping students make connections across curricula.

Preparing for the classroom
✓ Student learning needs
✓ Pedagogical knowledge
✓ Content knowledge
✓ Pedagogical content knowledge
✓ Victorian Curriculum F-10 / VEYLDF
✓ Practice Principles
✓ HITS

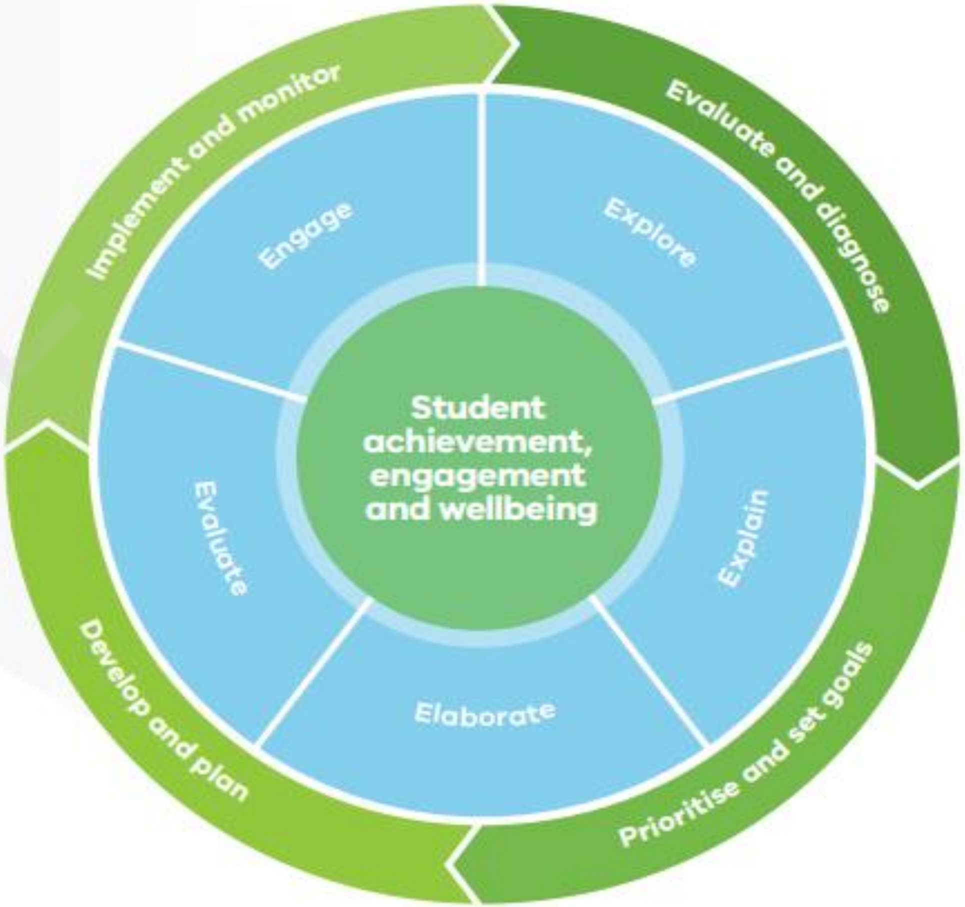
Reflecting on your practice
✓ Assess student learning progress and provide feedback
✓ Moderate student work with colleagues
✓ Use various types of evidence of student learning to inform the next planning cycle
✓ Undertake a reflection on your teaching practice using Diagnostic Tools for Practice Principles

Curriculum	Pedagogy	Assessment
defines what students should learn, and the associated progression or continuum of learning	describes how students will be taught and supported to learn	describes student progression in learning

Adapted from Victorian Curriculum F–10: Revised curriculum planning and reporting guidelines

- This higher education concept is distinct from the elementary and high school "integrated curriculum“ movement.
- Integrated studies involve bringing together traditionally separate subjects so that students can grasp a more authentic understanding.
- Interdisciplinary curricula has been shown by several studies to support students’ engagement and learning.
- Specifically integrating science with reading comprehension and writing lessons has been shown to improve students’ understanding in both science and English language arts.
- Reflective Approach
- Reflective teaching is a process where teachers think over their teaching practices , analyzing how something was taught and how the practice might be improved or changed for better learning outcomes.

- Some points of consideration in the reflection process might be what is currently being done, why it's being done and how well students are learning .
- We can use reflection as a way to simply learn more about our own practice, improve a certain practice (small groups and cooperative learning, for example) or to focus on a problem students are having.
- The Pedagogical Model describes what effective teachers do in their classrooms to engage students in intellectually challenging work.
- It provides an overview of the learning cycle and breaks it down into five domains or phases of instruction: Engage, Explore, Explain, Elaborate and Evaluate.



Constructive Approach	Reflective Approach	Collaborative Approach	Integrative Approach	Inquiry-Based Approach
• Student-centered • Builds on past knowledge • Instructor= Facilitator • Learning by doing • May be relatively slow-paced	• Continuous assessment of pedagogical practices • Model-approach for the trainee-teachers • Outcome-based • Involves timely reassessment of learning objectives	• Involves teamwork • Pools in different abilities of learners • Teachers and students may/may not work in teams • Enables the use of different teaching practices with different groups	• Integrates the classroom with the outside world • Makes the curriculum more realistic and relatable • Encourages application of acquired knowledge • Kindles interest in Mathematics and Science, specifically	• Student-centered • Excites curiosity in the learners • Enhances problem-solving skills • Four Types: 1. Confirmation 2. Structure 3. Guided 4. Open

- What are the Different Aspects of Pedagogy?
- According to Social pedagogy, education is crucial to a child's social development and it continues to support one's growth throughout his/her life.
- Thus, education and social aspects must go hand in hand, since students are social beings; they need education for effective communication.
- Culturally Responsive Pedagogy-
- In a culturally diverse society, teachers should adopt culturally responsive pedagogy strategies to meet the requirements of all students.
- The dimensions such as personal, institutional and instructional norms are put together to recognize the cultural differences among students.
- Good, modern-day pedagogy involves the interconnection of learning ideas, ways and methods of training or coaching them, and the achievement of results obtained following action after the learning.

Factors Affecting Pedagogy

- Learning Styles of Students
- Availability of Additional Resources
- Competence of the Instructor
- Educational Systems
- Field of Study

- Effective pedagogies are an amalgamation of various teaching techniques to improve higher-order thinking and cognitive abilities among students.
- Learning and pedagogy are well integrated and compliment each other in various ways, such as-
- Pedagogy is based on a student-centered approach; students can learn at their own pace and take full responsibility for learning.
- Pedagogy allows teachers to evaluate individual students' performance regularly; thus helps teachers to understand if a student is moving towards their target outcomes or not.
- Pedagogy allows students to meet like-minded people and learn from their peers.
- Pedagogy in education focuses on the evaluation, analysis, and compression that helps students develop cognitive skills.
- Pedagogy in the teaching sector can play a game-changer role.
- Studies have shown that pedagogy can support the high-level initiatives and resources required to prepare fearless learners for tomorrow.

Importance of Pedagogy in teaching:

- *Improved Quality of learning*
- *Students more receptive during learning sessions*
- *Improved student participation*
- *Knowledge imparted effectively across a spectrum of learners*
- *Development of higher cognitive skills in students.*

Engage	Explore	Explain	Elaborate	Evaluate
Teachers know their students well and engage them in building supportive, inclusive and stimulating learning environments. Teachers motivate and empower students to manage their own learning and develop agency.	Teachers present challenging tasks to support students in generating and investigating questions, gathering relevant information and developing ideas. They help students expand their perspectives and preconceptions, understand learning tasks and prepare to navigate their own learning.	Teachers explicitly teach relevant knowledge, concepts and skills in multiple ways to connect new and existing knowledge. They monitor student progress in learning and provide structured opportunities for practising new skills and developing agency.	Teachers challenge students to move from surface to deep learning, building student ability to transfer and generalise their learning. They support students to be reflective, questioning and self-monitoring learners.	Teachers use multiple forms of assessment and feedback to help students improve their learning and develop agency. They monitor student progress and analyse data to draw conclusions about the effectiveness of their teaching practices, identify areas for improvement and address student individual needs.

- Experiential Learning- Principles of EL
- Experiential learning occurs when carefully chosen experiences are supported by reflection, critical analysis and synthesis,
- Experiences are structured to require the student to take initiative, make decisions and be accountable for results;
- Throughout the experiential learning process, the student is actively engaged in posing questions, investigating, experimenting, being curious, solving problems, assuming responsibility, being creative and constructing meaning
- Students are engaged intellectually, emotionally, socially, soulfully and/or physically.
- This involvement produces a perception that the learning task is authentic,
- The results of the learning are personal and form the basis for future experience and learning
- Relationships are developed and nurtured: student to self, student to others and student to the world at large

Lecture: 37

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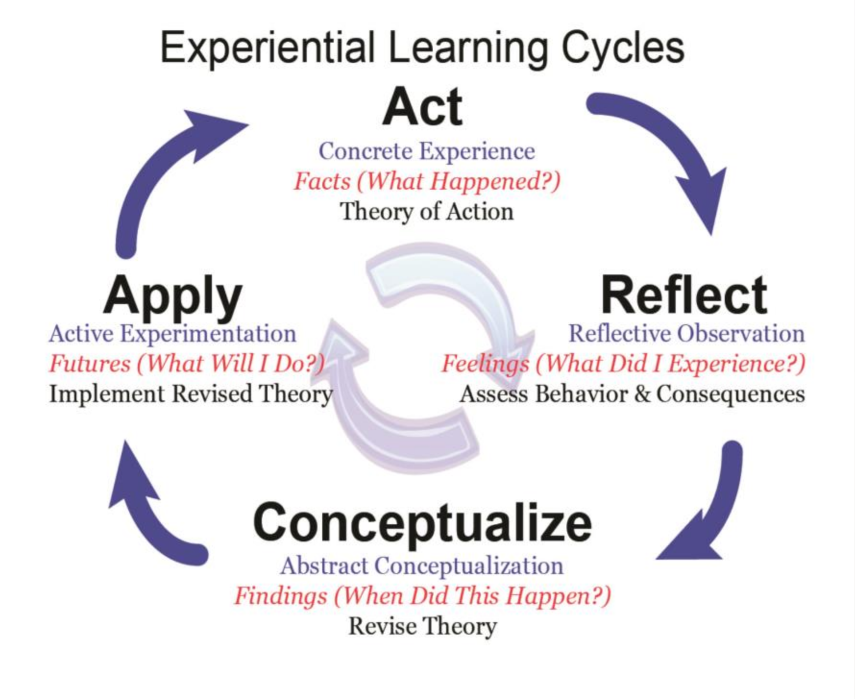
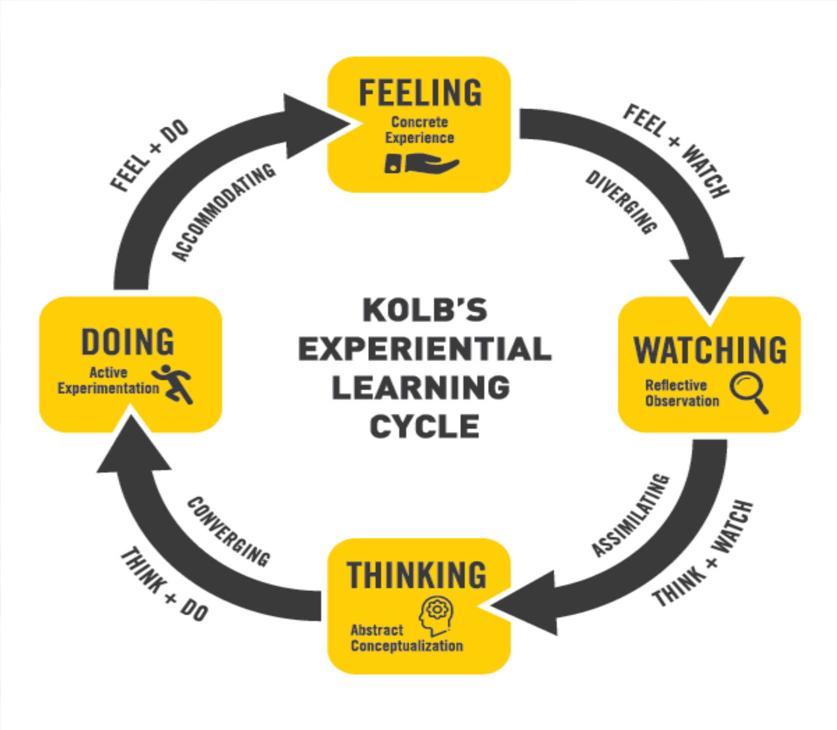
- The instructor and student may experience success, failure, adventure, risk-taking and uncertainty, because the outcomes of the experience cannot totally be predicted
- Opportunities are nurtured for students and instructors to explore and examine their own values
- The instructor's primary roles include setting suitable experiences, posing problems, setting boundaries, supporting students, insuring physical and emotional safety, and facilitating the learning process,
- The instructor recognizes and encourages spontaneous opportunities for learning
- Instructors strive to be aware of their biases, judgments and pre-conceptions, and how these influence the student
- The design of the learning experience includes the possibility to learn from natural consequences, mistakes and successes.

- The Experiential Learning Process
- Although learning content is important, learning from the process is at the heart of experiential learning -
- Experiencing/Exploring “Doing” -
- Students will perform or do a hands-on minds-on experience with little or no help from the instructor
- Sharing/Reflecting “What Happened?” -
- Students will share the results, reactions and observations with their peers
- Processing/Analyzing “What’s Important?” -
- Students will discuss, analyze and reflect upon the experience
- Generalizing “So What?” -
- Students will connect the experience with real world examples, find trends or common truths in the experience, and identify “real life” principles that emerged
- Application “Now What?” -
- Students will apply what they learned in the experience (and what they learned from past experiences and practice) to a similar or different situation

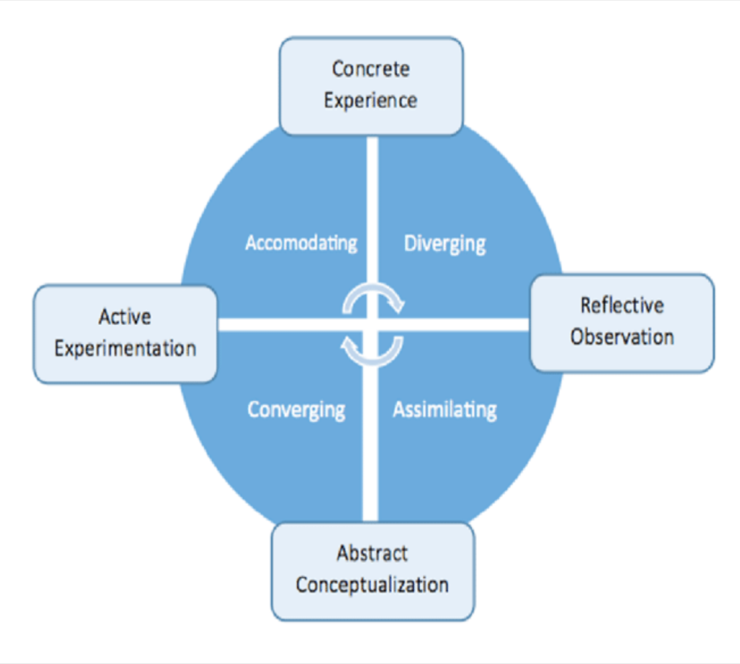


- Qualities of experiential learning are those in which students decide themselves to be personally involved in the learning experience…
- Student Roles in Experiential Learning -
- 1. Students will be involved in problems which are practical, social and personal
- 2. Students will be allowed freedom in the classroom as long as they make headway in the learning process
- 3. Students often will need to be involved with difficult and challenging situations while discovering
- 4. Students will self-evaluate their own progression or success in the learning process which becomes the primary means of assessment
- Students will learn from the learning process and become open to change;
- this change includes less reliance on the instructor and more on fellow peers, the development of skills to investigate (research) and learn from an authentic experience, and the ability to objectively self-evaluate one's performance
- Experiential Learning Opportunities in Higher Education
- Apprenticeship Experiences
- Clinical Experiences - hands-on experiences

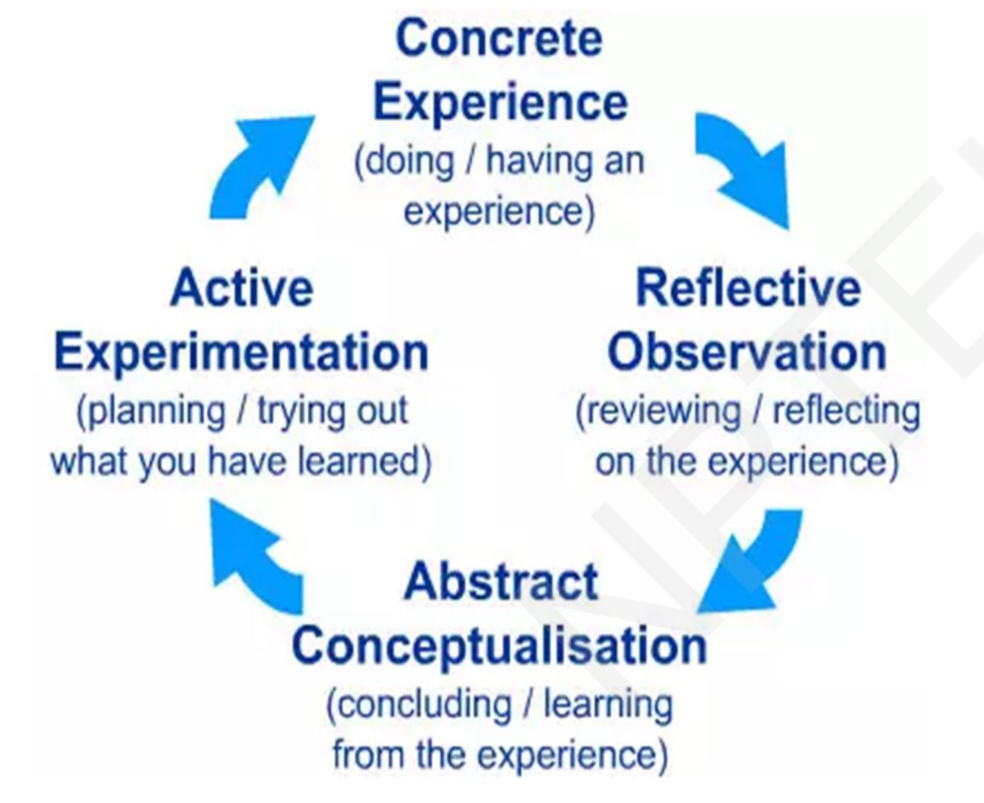
- Cooperative Education Experiences -paid professional work experiences
- Fellowship Experiences
- Field Work Experiences
- Internship Experiences are job-related
- Practicum Experiences are often a required component of a course
- Service Learning Experiences -performing a job within the community
- Student Teaching Experiences -on-site experience in a partner school
- Study Abroad Experiences
- Volunteer Experiences
- The **Experiential Learning Cycle-David Kolb's** experiential learning style theory is typically represented by a four-stage learning cycle in which the learner 'touches all the bases'



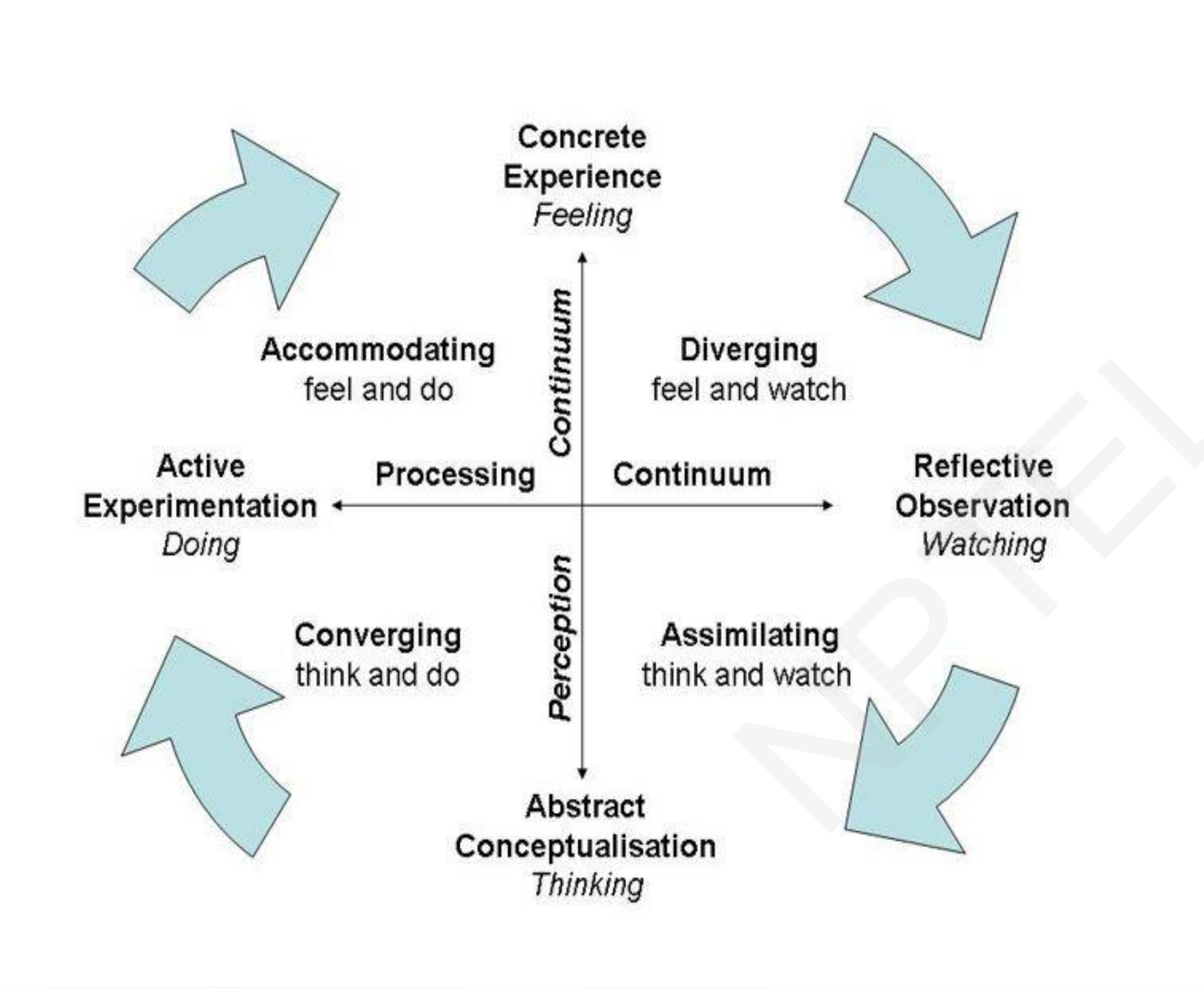
- 1. **Concrete Experience** - a new experience or situation is encountered, or a reinterpretation of existing experience
- 2. **Reflective Observation** of the New Experience - of particular importance are any inconsistencies between experience and understanding
- 3. **Abstract Conceptualization-** reflection gives rise to a new idea, or a modification of an existing abstract concept (the person has learned from their experience)
- 4. **Active Experimentation** - the learner applies their idea(s) to the world around them to see what happens
- Effective learning -when a person progresses through a cycle of four stages: of
- (1) having a concrete experience followed by (2) observation of and reflection on that experience which leads to (3) the formation of abstract concepts (analysis) and generalizations (conclusions) which are then (4) used to test hypothesis in future situations, resulting in new experiences

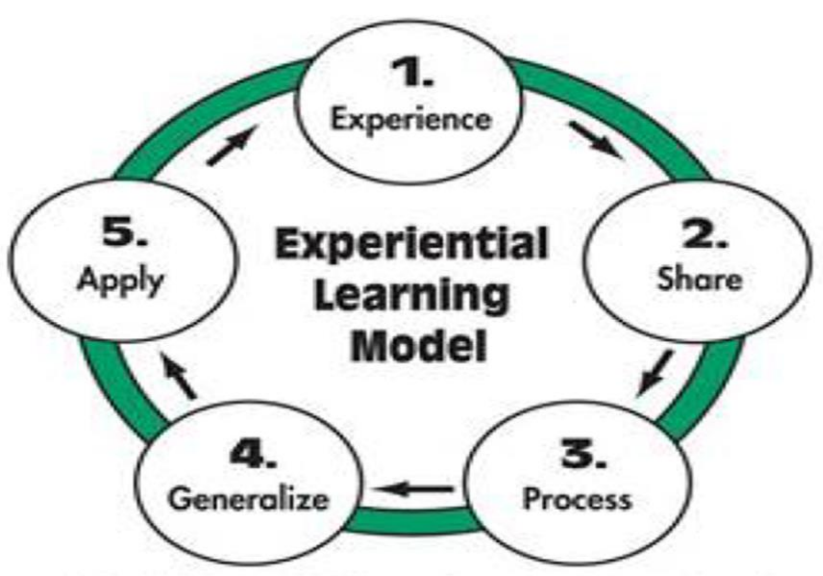


The experiential theory proposed by Kolb takes a more holistic approach and emphasizes how experiences, including cognition, environmental factors, and emotions, influence the learning process.



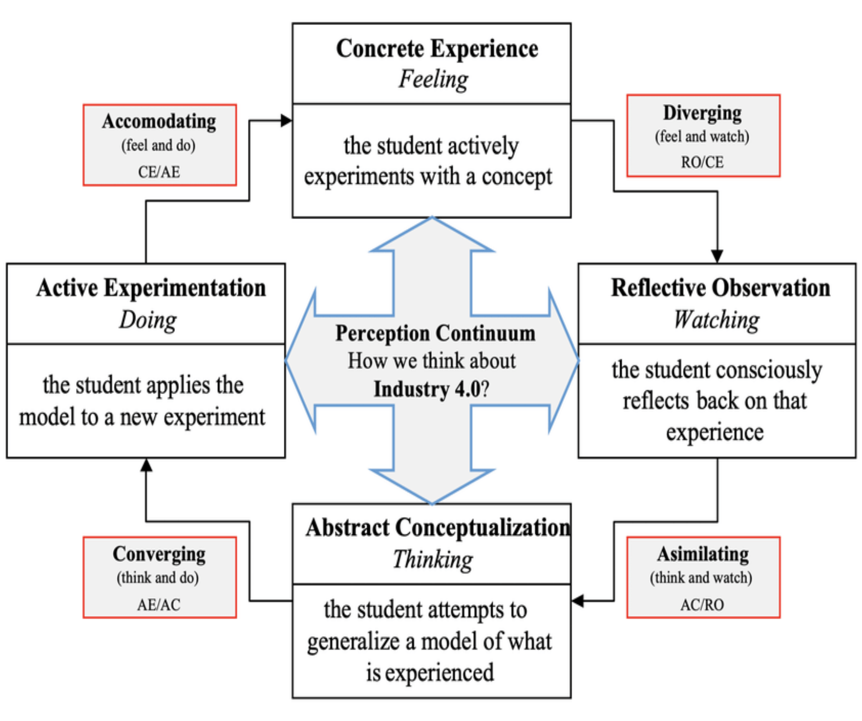
Kolb (1984) views learning as an integrated process with each stage being mutually supportive of and feeding into the next



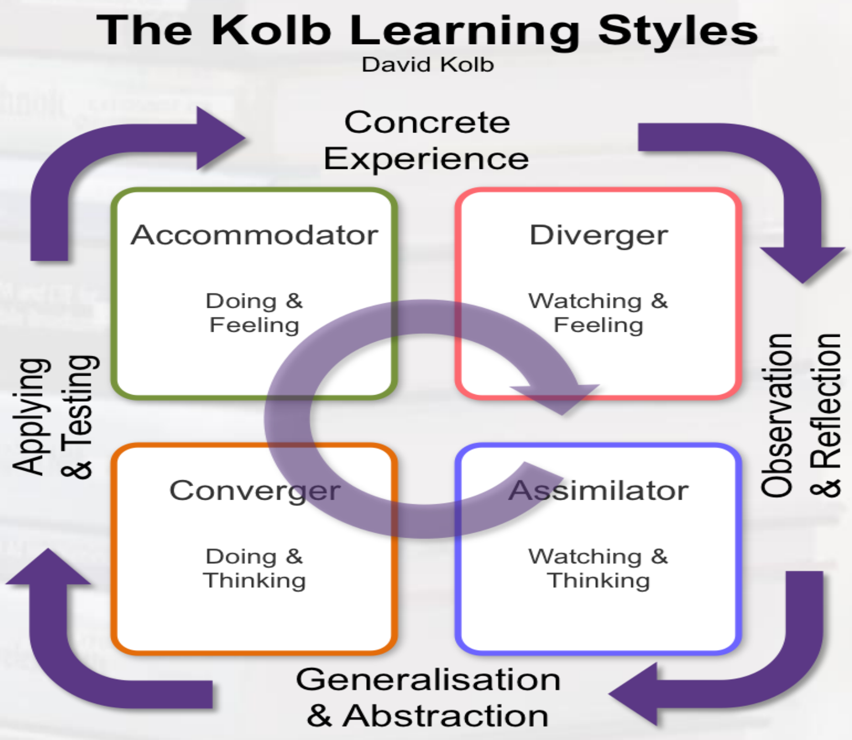


Constructivist learning theory
(Brooks & Brooks, 1993)

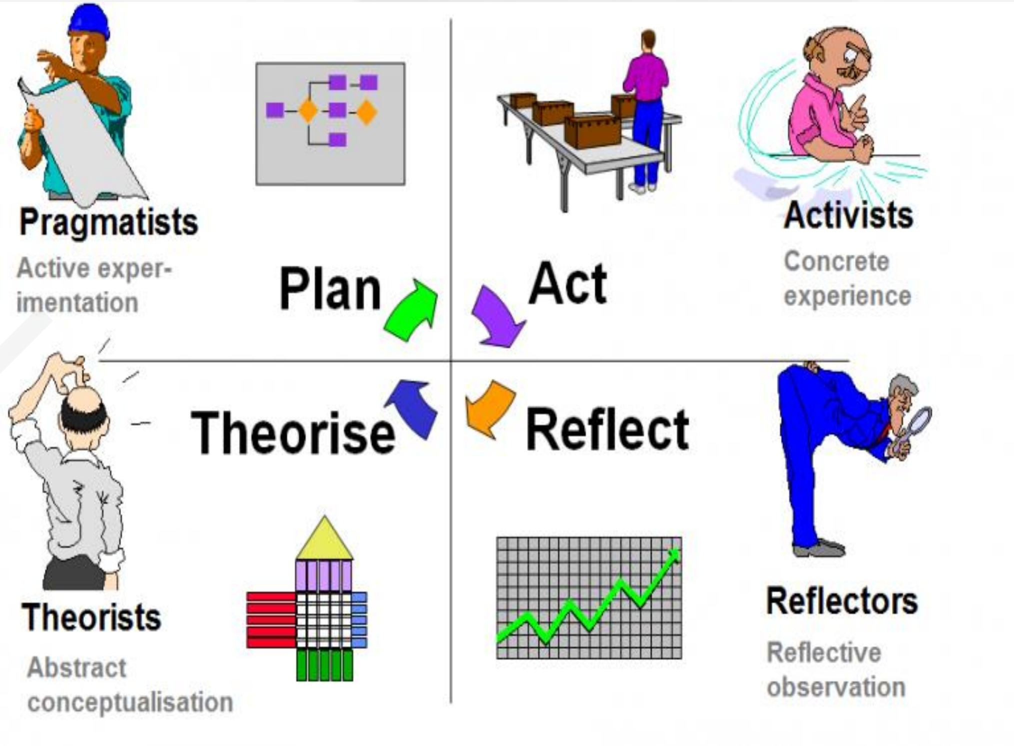
- The experiential learning model is a cyclical process of learning experiences.
- For effective learning to occur, the learner must complete the entire cycle.
- The 4 stages of the model are dissected by two continuums relating to perception and processing.
- Perception: The model explains how we can perceive new information in different ways, either more through thinking or by feeling.
- By immersing ourselves in rich environments, we experience the concrete, tangible, felt qualities of the world through our senses.
- We can also experience events more through thinking than feeling, where we grasp new information through models, representations, or logical analysis.
- Processing: Processing relates to how we approach a task, and whether we learn by watching or by doing.
- Through reflective observation we observe an environment from a distance and often through multiple perspectives as we search for an understanding of the situation at hand.



- Active experimentation is far more concerned with taking action, and learning through in-the-moment experiences,
- Kolb also described 4 learning styles, of which each of us broadly aligns to one.
- Diverging: These individuals prefer to look at things from multiple perspectives.
- They can be sensitive to stimuli in the environment, and prefer to watch rather than do.
- They are great at gathering information and using their imagination.
- Assimilating: These individuals prefer a concise, logical approach.
- Ideas and concepts are more important than people.
- They are great at taking in lots of information and organizing it in a clear and logical way.



- Converging: These individuals prefer technical tasks and like to be left alone to solve them.
- Interpersonal skills are not as important to them.
- They are great at solving problems and finding practical uses for ideas.
- Accommodating: These individuals are hands-on learners, and rely on intuition over logic.
- They can often act on "gut feeling" and will rely on others analysis rather than their own.
- They are great at bringing forward new challenges and challenging others.
- Assessing each of the learning styles is useful at the individual level in understanding your preferences for learning, which malleable across different disciplines.
- What is great about well-designed experiential learning scenarios is that every type of individuals can be accounted for.



- PEDAGOGIES THAT FACILITATE EXPERIENTIAL LEARNING
- Internships meet most criteria easily: participative, interactive, contact with environment, and variability/uncertainty
- Computer-Assisted Instruction-may well be applied, since its focus is on content
- Live Case- An approach which meets the criteria well is the live case approach
- Business Gaming -& Simulated experiential Learning
- Reality check survey- & Feedback
- https://www.youtube.com/watch?v=1vhYTKg3xK0
- https://www.youtube.com/watch?v=iUkZS0RiKpo
- https://www.youtube.com/watch?v=Rp-gaV-uSIo
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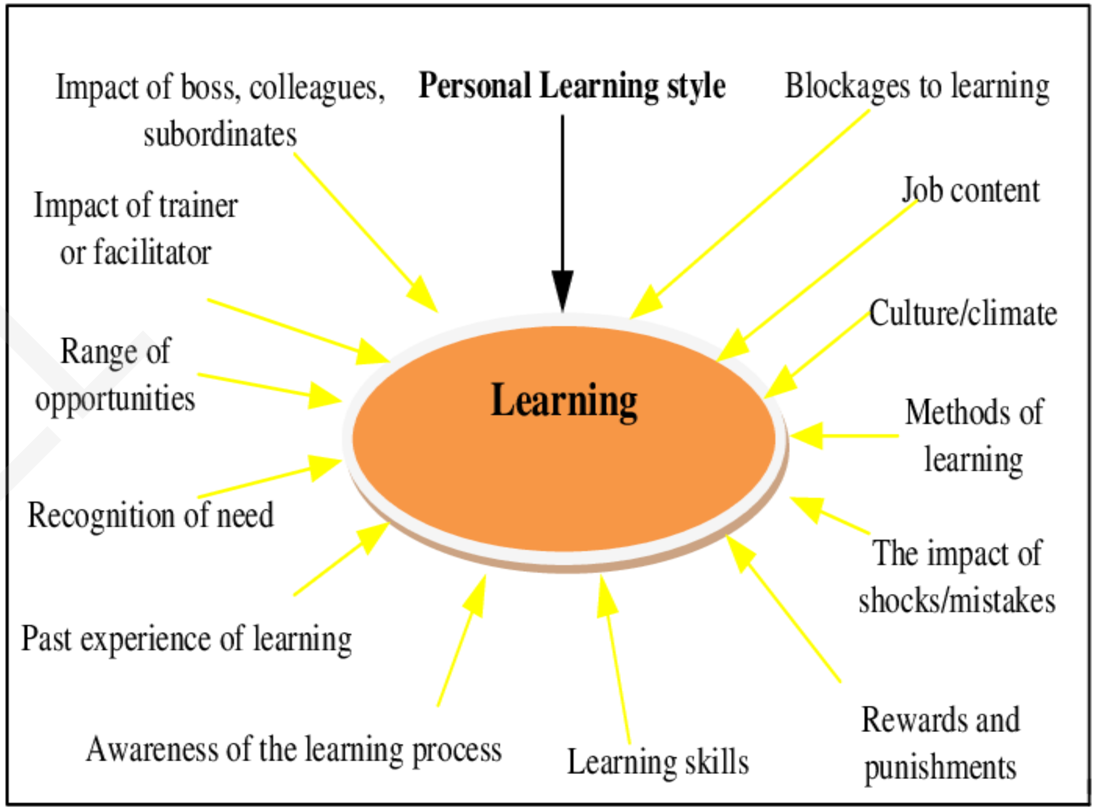
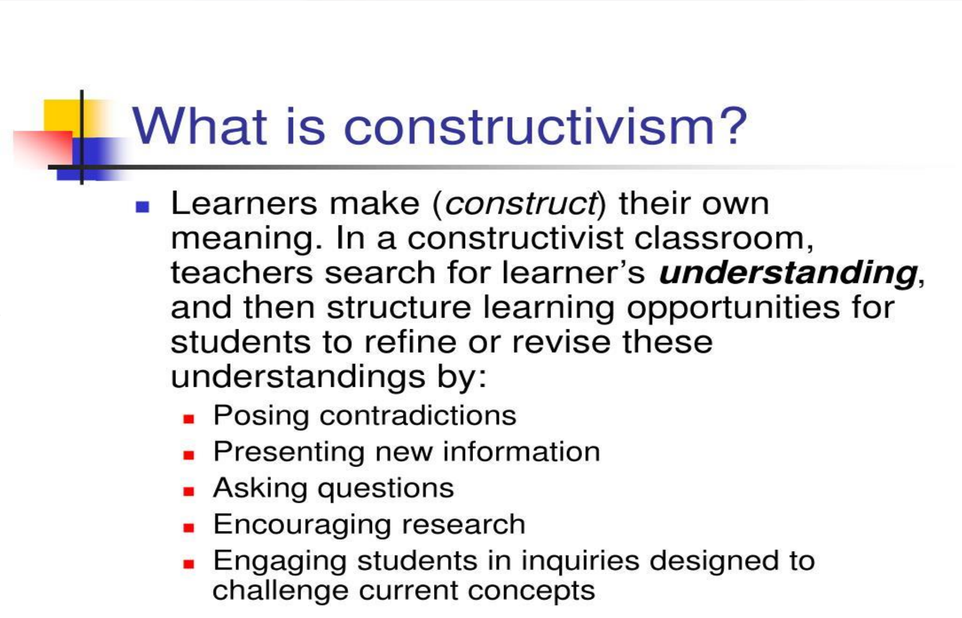
- Factors That Influence Learning Styles
- Kolb suggests that a number of different factors can influence preferred learning styles.
- 1 Some of the factors that he has identified include:
- Adaptive competencies
- Career choice
- Current job role
- Educational specialization
- Personality type
- The experiential theory proposed by Kolb takes a more holistic approach and emphasizes how experiences, including cognition, environmental factors, and emotions, influence the learning process.



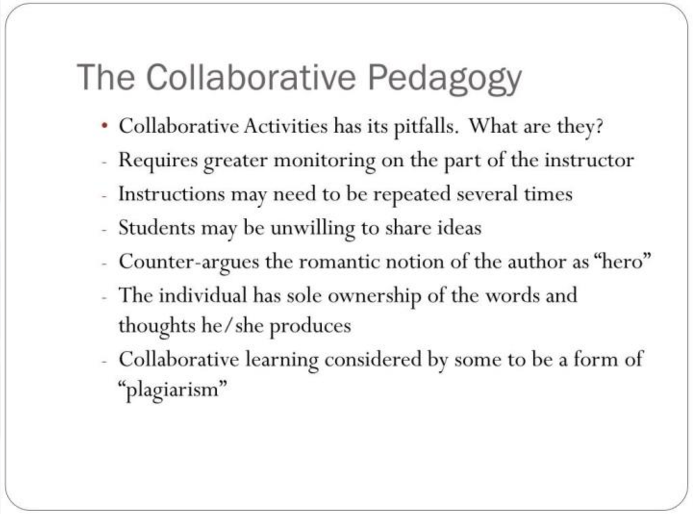
- Constructivist pedagogy- CP
- Constructivist pedagogy is where instructors encourage students to construct their own knowledge through experiences and activities versus being lectured on abstract concepts.
- Educators who teach using a constructivist pedagogy promote skills and subject mastery through hands-on lessons and self-guided learning.
- Students are asked to form their own conclusions about a topic through their own discoveries.
- It refers to a theory of instruction where students are asked to absorb and understand course concepts through rich exploration.
- Constructivist pedagogy supports experiential learning, where students learn through doing.
- Put to practice, a non-constructivist approach may involve students reading a textbook chapter and answering discussion questions.
- A teacher using the constructivist method may ask students to complete a simulation conveying the stages of mitosis to engage students further.

- Ultimately, the constructivist pedagogy supports the idea that knowledge can't be passively absorbed and instead is constructed, or modified, through individual experiences.
- Constructivist teaching strategies help students understand the meaning of their learning materials, instead of just passively ingesting content.
- Rather than focusing on the subject or lesson being taught, educators are encouraged to focus on how the student learns.
- Inquiry-based learning encourages students to ask questions and complete research while learning various concepts.
- The pedagogy focuses on helping learners acquire the skills necessary to develop their own ideas, as well as question themselves and group members in a constructive way.
- The four steps of inquiry-based learning are:
- Developing problem statements that require students to pitch their question using a constructed response, further inquiry and citation.



- Researching the topic using time in class where the instructor can guide students in their learnings
- Presenting what they've learned to their peers or to a small group
- Asking students to reflect on what worked about the process and what didn't.
- Students focus on how they learned in addition to what they learned, to activate metacognition skills (or thinking about thinking).
- Collaborative pedagogy- is a type of learning theory where students work with their peers to find solutions to problems.
- Using peer instruction, students may teach one another and, in the process, clarify their own misconceptions.
- Interpersonal engagement and peer-to-peer interaction are at the heart of collaborative pedagogy.
- Collaborative pedagogy can help promote a greater sense of belonging among students, expose students to diverse points of view and equip learners with the soft skills needed for their careers.

- It refers to the idea that the best learning happens in small groups or pairs.
- As opposed to absorbing information in isolation, students work together to solve complex problems or complete learning activities.
- Educators step aside and let partners or groups self-direct their own learning process.
- Put to practice, the collaborative pedagogy model may involve students developing a business strategy for an organization of their choosing.
- It rejects the notion that students can think, learn and write effectively in isolation.
- This is a learner-centered strategy that strives to maximize critical thinking, learning and writing skills through peer-to-peer interaction and interpersonal engagement.



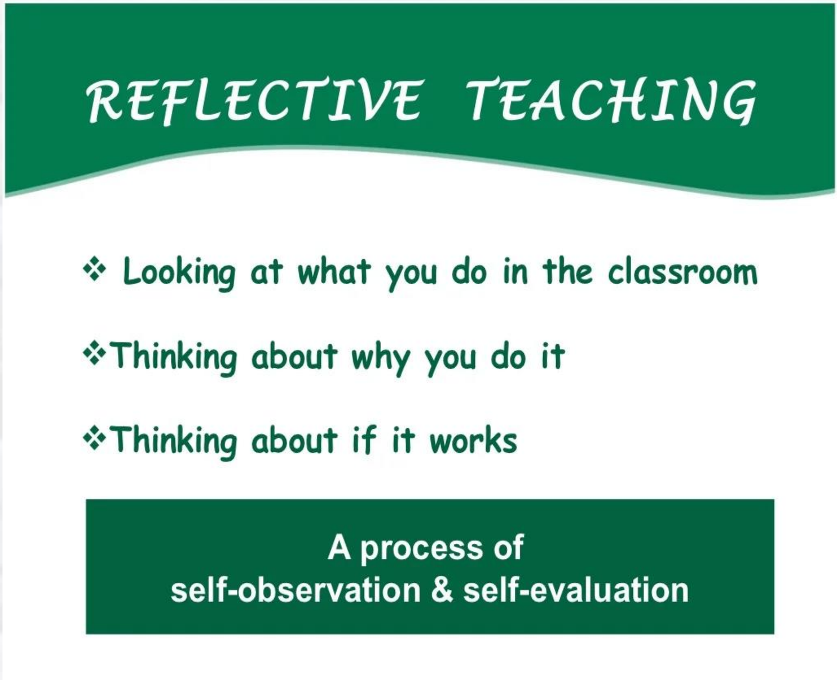
The Collaborative Pedagogy

- Collaborative Activities has its pitfalls. What are they?
- Requires greater monitoring on the part of the instructor
- Instructions may need to be repeated several times
- Students may be unwilling to share ideas
- Counter-argues the romantic notion of the author as "hero"
- The individual has sole ownership of the words and thoughts he/she produces
- Collaborative learning considered by some to be a form of "plagiarism"

- Integrative learning is a teaching pedagogy that helps students connect concepts to real-world experiences.
- Students build upon soft skills including communication and critical thinking, which they can then apply to a variety of issues or scenarios.
- Integrative learning can help students adapt to the needs of a rapidly changing workplace and can help learners develop the transferable skills needed to thrive post-graduation.
- Within the classroom, integrative learning can help students develop a deeper understanding and appreciation for their subject area.
- It refers to a method of teaching that helps bridge the gap between the classroom and workplace.
- Students are asked to form connections between what they're learning and how to apply those learnings to society and their future careers.

- Integrative learning often involves three steps including integrative inquiry (asking meaningful questions), application and transfer (applying skills to new situations) and reflection (making personal and professional plans based on self-reflection).
- **An example of integrative learning is an e-Portfolio,** where students are asked to select evidence of their learning and reflect on their knowledge gained and how to apply their understanding to academia and the real world.
- It is the process of making connections between concepts and experiences so that information and skills can be applied to novel and complex issues or challenges.
- Reflective pedagogy- is a teaching model where educators continually reflect upon their lessons and curriculum to improve future iterations of their course.

- Instructors who use the reflective pedagogy model regularly gather data on student satisfaction, engagement and belonging to help inform subsequent lessons.
- They typically poll learners before, during and after class to ensure their course is taught in a way that's conducive to student success.
- Reflective pedagogy refers to a curriculum approach where educators collect and analyze evidence of effective teaching.
- Educators typically turn to evidence, including qualitative and quantitative data from students, colleagues' perceptions, personal experiences and educational theory and research.
- Self-assessment is a core component of reflective pedagogy. Here, educators may record themselves giving a lecture and then observe their behavior and delivery to critique the efficacy of their lessons.
- Educators may also use a teaching inventory to assess their current teaching practices and spark new ideas for future lessons and/or courses.



- Thus, it is the process of making connections between concepts and experiences so that information and skills can be applied to novel and complex issues or challenges.
- Reflective pedagogy -encourages the instructor to reflect upon lessons, projects and assessments, with the goal of improving them for future use.
- Students are also encouraged to reflect on their performance on assessments and look for areas where they can improve.
- Reflective pedagogy is a teaching model where educators continually reflect upon their lessons and curriculum to improve future iterations of their course.
- Instructors who use the reflective pedagogy model regularly gather data on student satisfaction, engagement and belonging to help inform subsequent lessons.

The 4Rs Model of Reflective Thinking

Level	Stage	Questions to get you started
1	Reporting and Responding	Report what happened or what the issue or incident involved. Why is it relevant? Respond to the incident or issue by making observations, expressing your opinion, or asking questions.
2	Relating	Relate or make a connection between the incident or issue and your own skills, professional experience, or discipline knowledge. Have I seen this before? Were the conditions the same or different? Do I have the skills and knowledge to deal with this? Explain.
3	Reasoning	Highlight in detail significant factors underlying the incident or issue. Explain and show why they are important to an understanding of the incident or issue. Refer to relevant theory and literature to support your reasoning. Consider different perspectives. How would a knowledgeable person perceive/ handle this? What are the ethics involved?
4	Reconstructing	Reframe or reconstruct future practice or professional understanding. How would I deal with this next time? What might work and why? Are there different options? What might happen if...? Are my ideas supported by theory? Can I make changes to benefit others?

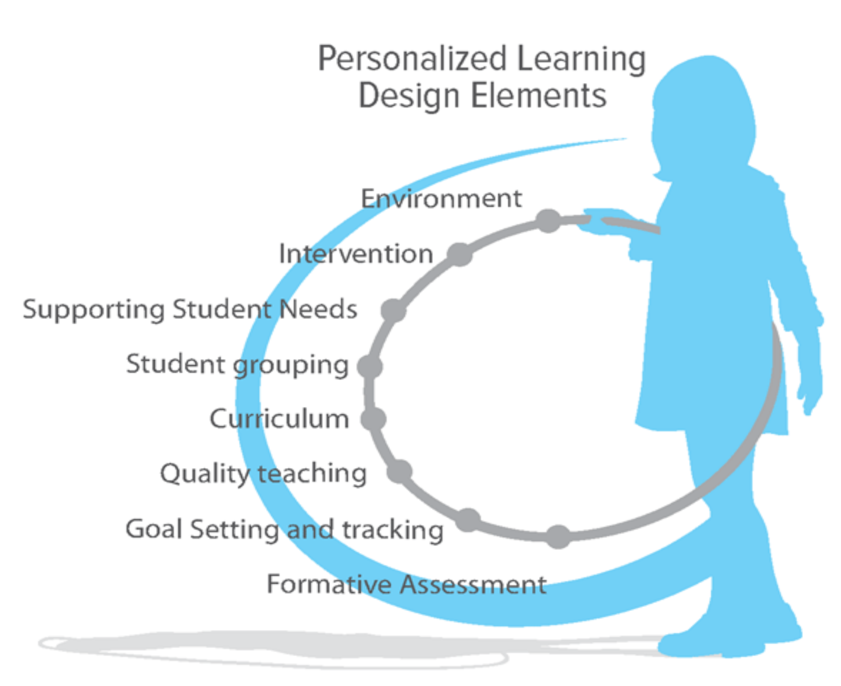
- Reflective pedagogy refers to a curriculum approach where educators collect and analyze evidence of effective teaching.
- Self-assessment is a core component of reflective pedagogy.
- Here, educators may record themselves giving a lecture and then observe their behavior and delivery to critique the efficacy of their lessons.
- Educators may also use a teaching inventory to assess their current teaching practices and spark new ideas for future lessons and/or courses.
- **Critical pedagogy** asserts that issues of social justice and democracy are not distinct from acts of teaching and learning.
- It is a theory and practice that helps students question and challenge prevalent beliefs and practices—and achieve critical consciousness

Critical pedagogy questions

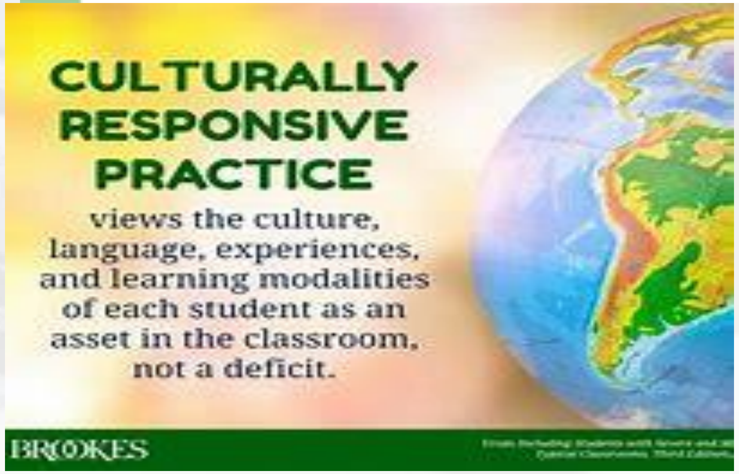
- Whose interests does certain types of knowledge serve?
- Who gets excluded as a result?
- Who is marginalized?
- What is the relationship between social class and knowledge taught in school?
- Why do we value scientific knowledge over informal knowledge?
- Why do teachers have to teach using “standard English”?
- How does school knowledge reinforce stereotypes about disadvantaged peoples?

- Critical pedagogy is a philosophy of education and social movement that involves questioning normalized power structures and hierarchies that exist today.
- There are several steps within critical pedagogy: unlearning, learning, relearning, reflection and evaluation.
- Critical pedagogy supports a student-centered classroom, where students are no longer complacent figures, but rather critical thinkers and active participants of their learning.
- Put to practice, Philosophy educators who have adopted critical pedagogy may ask students to critique and evaluate both sides of an argument when writing a paper—not just the side that the student favors.
- Critical pedagogy is a theory of learning that asks students to critique power structures in society.
- Teachers who adopt critical pedagogy regularly challenge inequities that exist in families, health, education and more societal structures.

- It involves challenging dominant and familiar narratives that have become normalized in society.
- Students become their own decision makers and are able to rewrite their own experiences and perceptions of the world around them.
- In recent years, critical pedagogy has incorporated elements from fields including the Human and Civil Rights Movements, Indigenous rights movements, queer theory and feminist theory.
- Personalizing pedagogies-New applications of information technology have provided a variety of choices in higher education, not only about what is taught and learned, but also about how it is taught and learned.
- In recent years, there has been a lot of excitement about new ways to use information technology to meet the needs of learners more effectively, including new pedagogical techniques in individualization, learner-centeredness, and anytime-anywhere education.

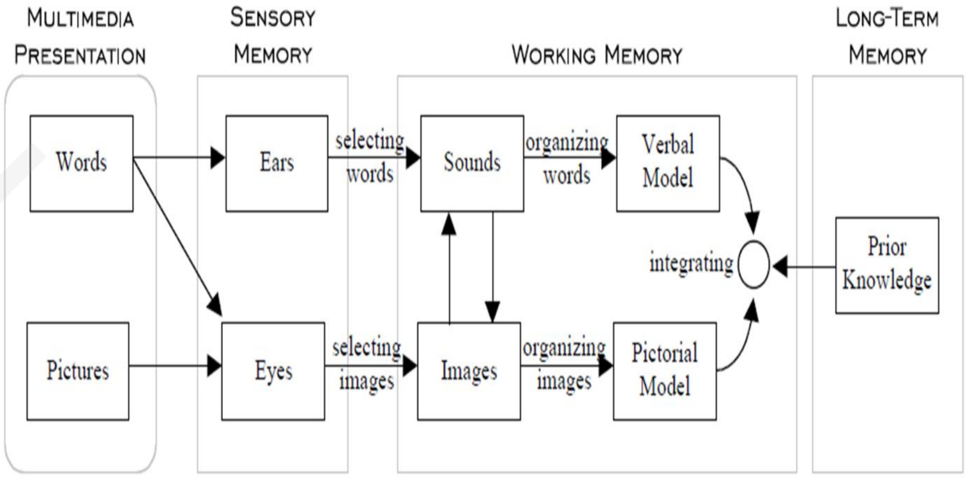


- The increase in non-formal, self-directed learning methods means that students have more access to information than ever before.
- It makes it easier for educators to track their learning through digital activities.
- But it also requires more attention in guiding them to the right sources, adjusting lecture content and adopting approaches purpose-built for engagement and collaboration.
- In many innovative pedagogies, there’s a power shared between educator and student.
- Students learn more independently, instead of following a set course of lectures and textbooks from an instructor.
- In many cases, students thrive in self-directed learning methods, while educators can use lecture time more effectively for discussion and collaborative work.
- Culturally responsive teaching is a more modern pedagogy that acknowledges, responds to and celebrates fundamental cultures.
- It strives to offer equitable access to education for students from all cultures.



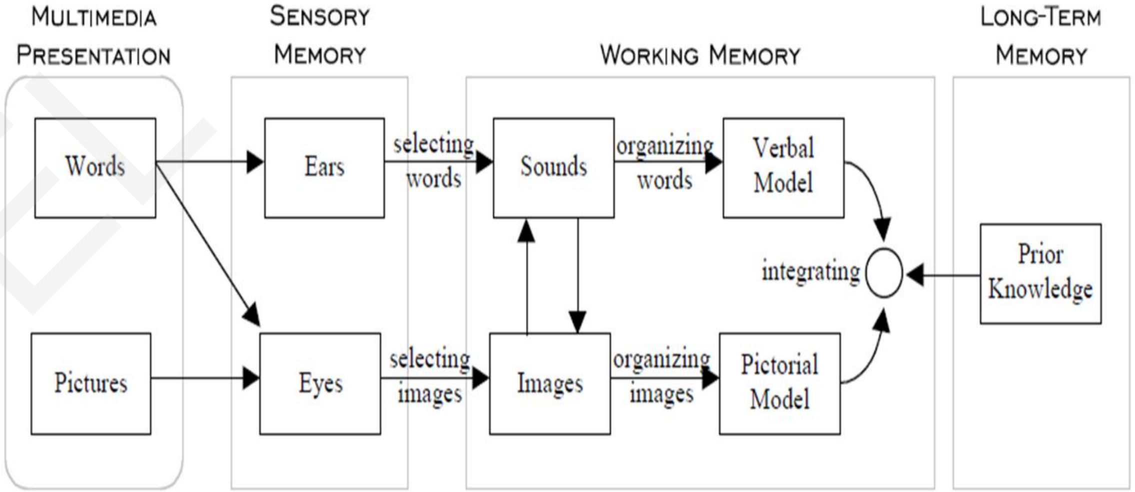
- Multimedia Learning-
- People learn better when multimedia messages are designed in ways that are consistent with how the human mind works and with research-based principles
- Multimedia provides a technology based constructivist learning environment where students are able to solve a problem by means of self explorations, collaboration and active participation,
- This model is based upon three primary assumptions (Mayer, 2001):
- 1. Visual and auditory experiences/information are processed through separate and distinct information processing 'channels'
- 2. Each information processing channel is limited in its ability to process experience/information
- 3. Processing experience/information in channels is an active cognitive process designed to construct coherent mental representations

Mayer bases his *cognitive theory of multimedia learning* on the following model.



- Multimedia- Presenting words (such as printed text or spoken text) and pictures (such as illustrations, photos, animation, or video)
- Multimedia learning-Building mental representations from words and pictures
- Multimedia instruction -Presenting words and pictures that are intended to promote learning

Mayer bases his *cognitive theory of multimedia learning* on the following model.



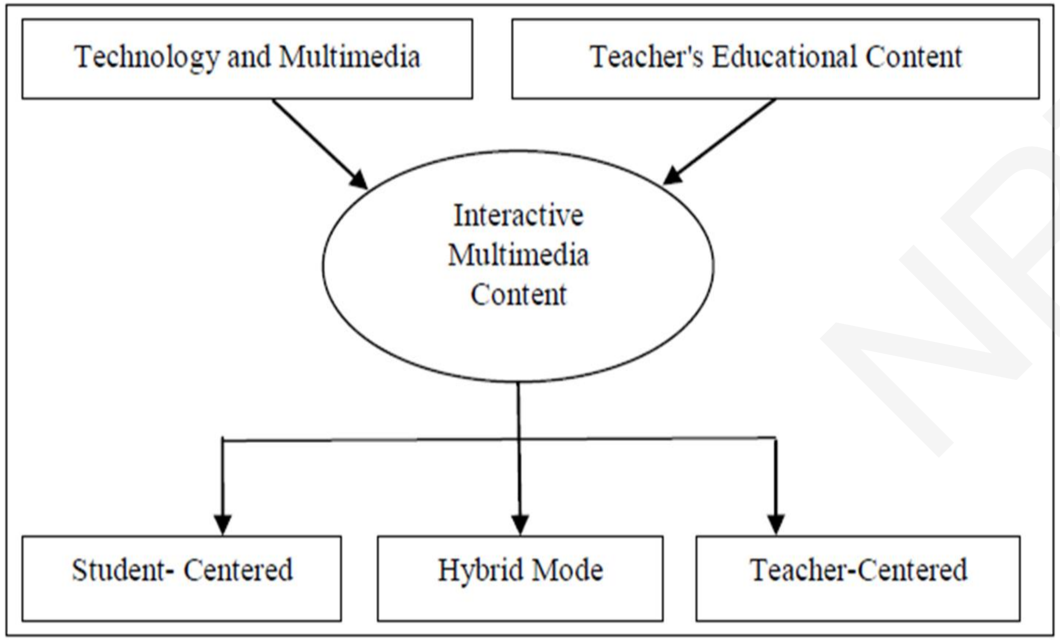


Fig.1using multimedia to represent content and delivering via various methods

- This model is activated through five steps:
- 1. (a) selecting relevant words for processing in verbal working memory,
- 2. (b) selecting relevant images for processing in visual working memory,
- 3. (c) organization selected words into a verbal mental model
- 4. (d) organizing selected images into a visual mental model, and
- 5. (e) integrating verbal and visual representations as well as prior knowledge" (Mayer, 2001).

Two Approaches to Multimedia Design

Approach	Starting point	Goal	Issues
Technology-centered	Capabilities of multimedia technology	Provide access to information	How can we use cutting edge technology in designing multimedia presentations?
Learner-centered	How the human mind works	Aid to human cognition	How can we adapt multimedia technology to aid human cognition?

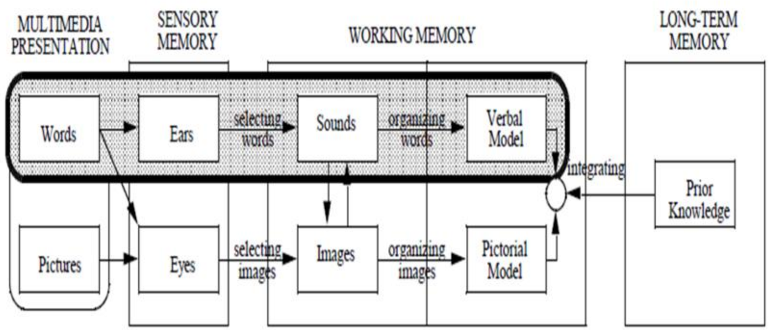
Two Goals of Multimedia Learning

Goal	Definition	Test	Example test item
Remembering	Ability to reproduce or recognize presented material	Retention	Write down all you can remember from the passage you just read.
Understanding	Ability to use presented material in novel situations	Transfer	List some ways to improve the reliability of the device you just read about.

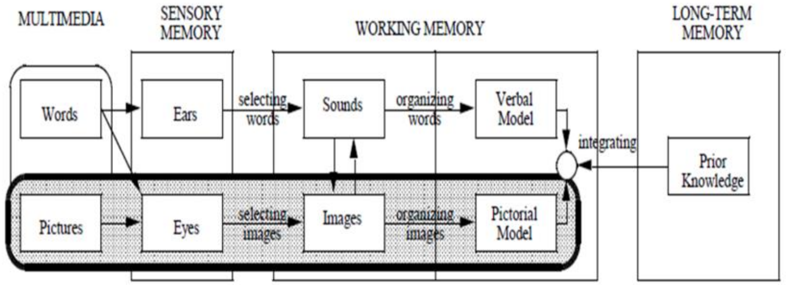
Three Assumptions of a Cognitive Theory of Multimedia Learning

Assumption	Description
Dual channels	Humans possess separate channels for processing visual and auditory information.
Limited capacity	Humans are limited in the amount of information that they can process in each channel at one time.
Active processing	Humans engage in active learning by attending to relevant incoming information, organizing selected information into coherent mental representations, and integrating mental representations with other knowledge.

Auditory/Verbal Channel Highlighted



Visual/Pictorial Channel Highlighted



- Principles of Multimedia Learning & Instruction

Empirical Results	Practical Applications
Multimedia Principle: Students learn better from words and pictures than from words alone.	On screen animation, slide shows, and narratives should involve both written or oral text and still or moving pictures. Simple blocks of text or auditory only links are less effect than when this text or narration is coupled with visual images. (Sample example)
Spatial Contiguity Principle: Students learn better when corresponding words and pictures are presented near rather than far from each other on the page or screen.	When presenting coupled text and images, the text should be close to or embedded within the images. Placing text under an image (i.e., a caption) is sufficient, but placing the text within the image is more effective. (Sample example)
Temporal Contiguity Principle: Students learn better when corresponding words and pictures are presented simultaneously rather than successively.	When presenting coupled text and images, the text and images should be presented simultaneously. When animation and narration are both used, the animation and narration should coincide meaningfully. (Sample example)

- Principles Contd……..

Coherence Principle: Students learn better when extraneous words, pictures, and sounds are excluded rather than included.	Multimedia presentations should focus on clear and concise presentations. Presentations that add "bells and whistles" or extraneous information (e.g. to increase interest) impede student learning. (Sample example).
Modality Principle: Students learn better from animation and narration than from animation and on-screen text.	Multimedia presentations involving both words and pictures should be created using auditory or spoken words, rather than written text to accompany the pictures. (Sample example)
Redundancy Principle: Student learn better from animation and narration than from animation, narration, and on-screen text.	Multimedia presentations involving both words and pictures should present text either in written form, or in auditory form, but not in both. (Sample example)
Individual Differences Principles: Design effects are stronger for low-knowledge learners than for high-knowledge learns and for high spatial learners rather than from low spatial learners.	The aforementioned strategies are most effective for novices (e.g., low-knowledge learners) and visual learners (e.g., high-spatial learners). Well structured multimedia presentations should be created for they are most likely to help.

- Conclusions about - Design of Multimedia Learning
- 1. **Theory-based**- The design of multimedia messages should be based on a theory of how the human mind works
- 2. **Research-based**- The design of multimedia messages should be based on research findings,
- Mayer has based the majority of his multimedia work on an integration of-
- Sweller's cognitive load theory
- (Chandler & Sweller, 1991; Sweller, 1999),
- Pavio's dual-coding theory (Clark & Paivio, 1991; Paivio, 1986), &
- Baddeley's working memory model (1986, 1992, 1999)

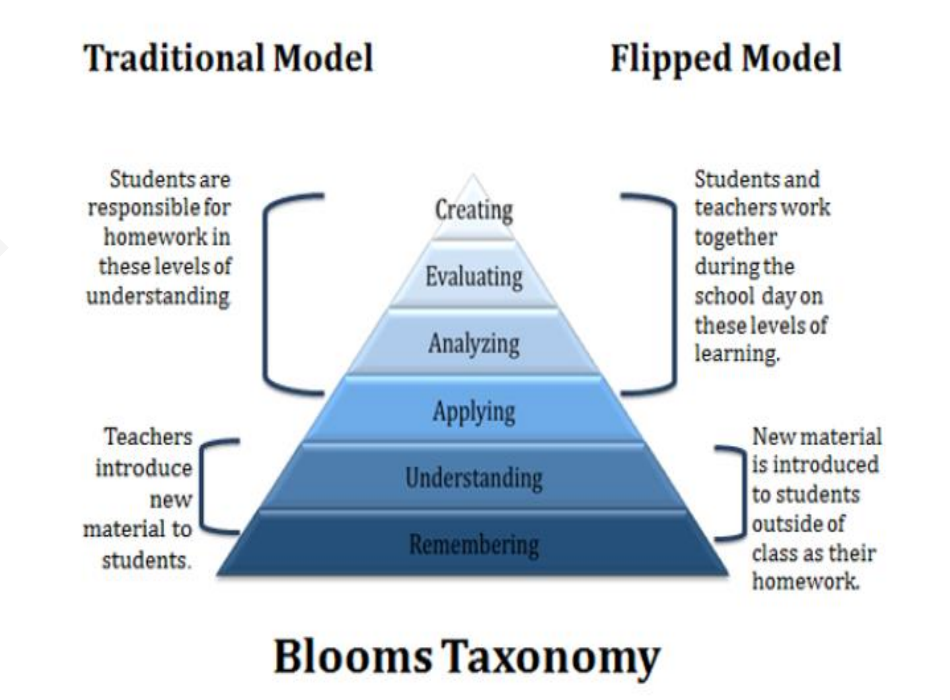
Three Kinds of Multimedia Learning Outcomes

Learning outcome	Cognitive description	Retention test score	Transfer test score
No learning	No knowledge	Poor	Poor
Rote learning	Fragmented knowledge	Good	Poor
Meaningful learning	Integrated knowledge	Good	Good

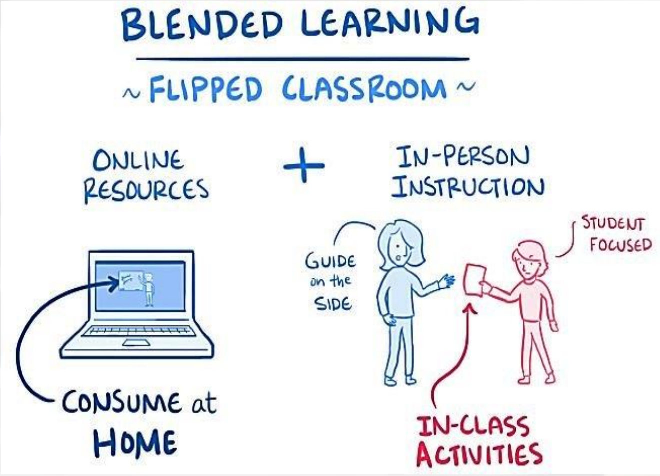
- Research-Based Principles for the Design of Multimedia Messages- Contd......
- Personalization principle: People learn better when the words are in conversational style rather than formal style
- Interactivity principle: People learn better when they have control over the pace of the presentation
- Signaling principle: People learn better when the words include cues about the organization of the presentation.
- Individual differences principle: Design effects are stronger for low knowledge learners than for high-knowledge learners.
- Design effects are stronger for high-spatial learners than for low-spatial learners.

- https://www.youtube.com/watch?v=0aq2P0DZqEI
- https://www.youtube.com/watch?v=Q5eY9k3v4mE
- https://www.youtube.com/watch?v=AJ3wSf-ccXo
- https://www.youtube.com/watch?v=p5i3f9E53Og

- **Blended Learning**
- Blended learning: Combines online educational materials and opportunities for interaction online with traditional place- based classroom methods.
- Requires the physical presence of both teacher and student, with
- some elements of student control over time, place, path, or place.
- **Benefits:**
- Convenience and Flexibility
- More comprehensive understanding of the course content.
- Social learning is supported
- **Pitfalls:**
- Target group specific content; distribution is a challenge
- Learners take time to adapt
- Assessment of performance is difficult



- **Blended-Flip Approach-**
- Flipped Classroom is the reverse of traditional learning
- Blended learning:
- Combines online educational materials and opportunities for interaction online with traditional place-based classroom methods
- Requires the physical presence of both teacher and student, with some elements of student control over time, place, path, or pace.
- Blended-Flip Approach
- Combination of blended and Flipped approach is named as blended-Flip teaching method
- Lack of resources compelled us to provide both the online and offline study materials inside the classroom.
- Student-student & student-teacher interaction took place inside the classroom.
- Students were allowed to ask questions to the teacher inside the classroom while going through the study materials.



- **Why & How of Blended-Flip Pedagogy-**
- Flexible Environment-

  Teacher creates flexible learning environment

  Opportunities for students to choose when & where they want to learn.

  Research shown that the blended-flipped classroom is able to improve students' cognitive flexibility & creative thinking skills.
- Learning culture

  More time to discuss in-depth about each topic

  Changes the traditional learning culture into learner centered class

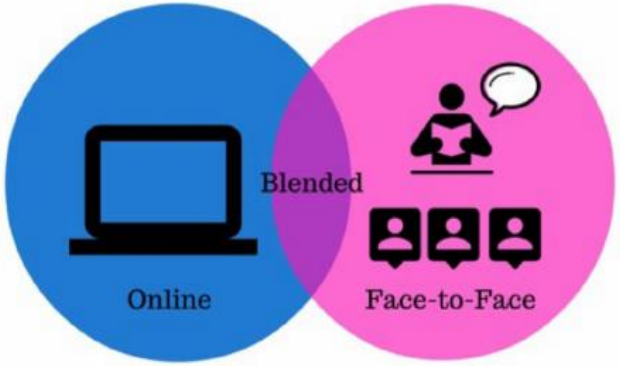
  Students are actively involved in knowledge construction
- Intentional content

  Teachers determine what they need to teach and which materials students should handle on their own.
- Professional educator

  More important in a flipped classroom than in traditional one.

  Flipped classroom model needs instructors who can observe

  Provide timely feedback continuously assess work and help students master content.



Blending the Best of Two Instructional Pedagogies

Figure 1 : Blended learning

Blended-Flip- Jonathan Bergmann and Aaron Sams pioneered 'BF Classroom' model of teaching in 2012 [1].

Pre-class activity	School computer lab was used for pre-class preparation Teacher was present to clarify students 'doubt. Students went through the study materials and video lessons in the presence of the teacher. Small groups were formed among the students to discuss the study material. Teacher presence ensured all students are preparing well before coming to the class.
In class activity	Open questions were put by the teacher Pair and share activities Student presentation and discussion Facilitator/Teacher moderates the session Individual/paired quiz Clarification of doubts
Post-class session	Evaluation of performance Rewarding the good scorers Intensive training for poor scorers Further doubt clarification

Blended Learning

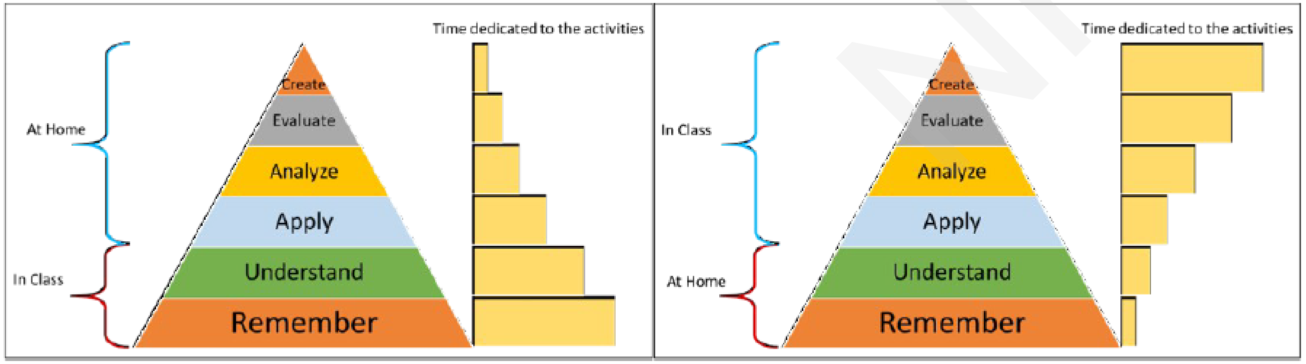


Figure 4 – Distribution of the time devoted to activities in the traditional class (left) and Flipped Classroom (right) based on Bergmann e Sams (2014, p. 30)

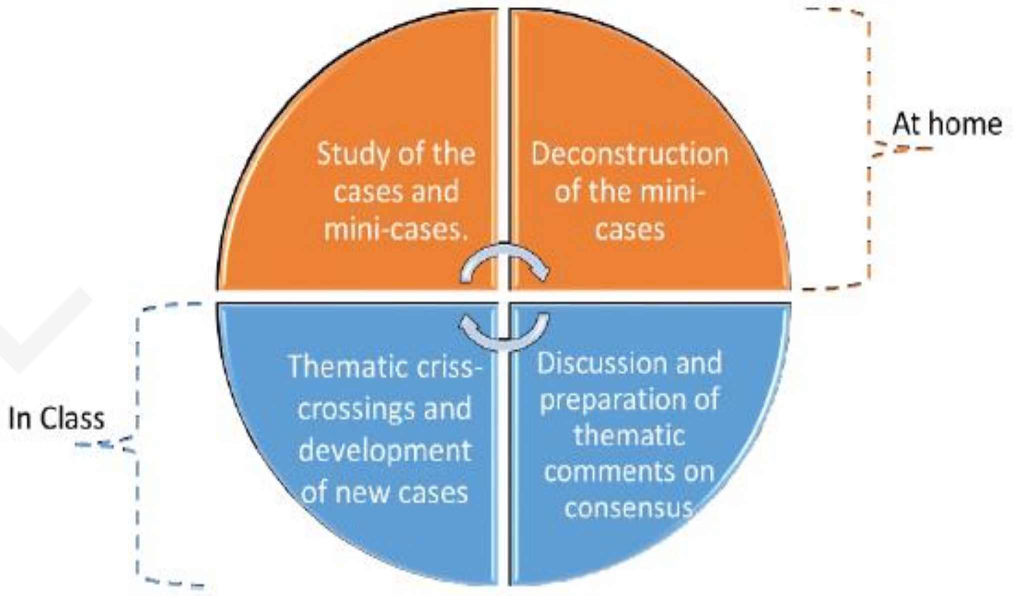
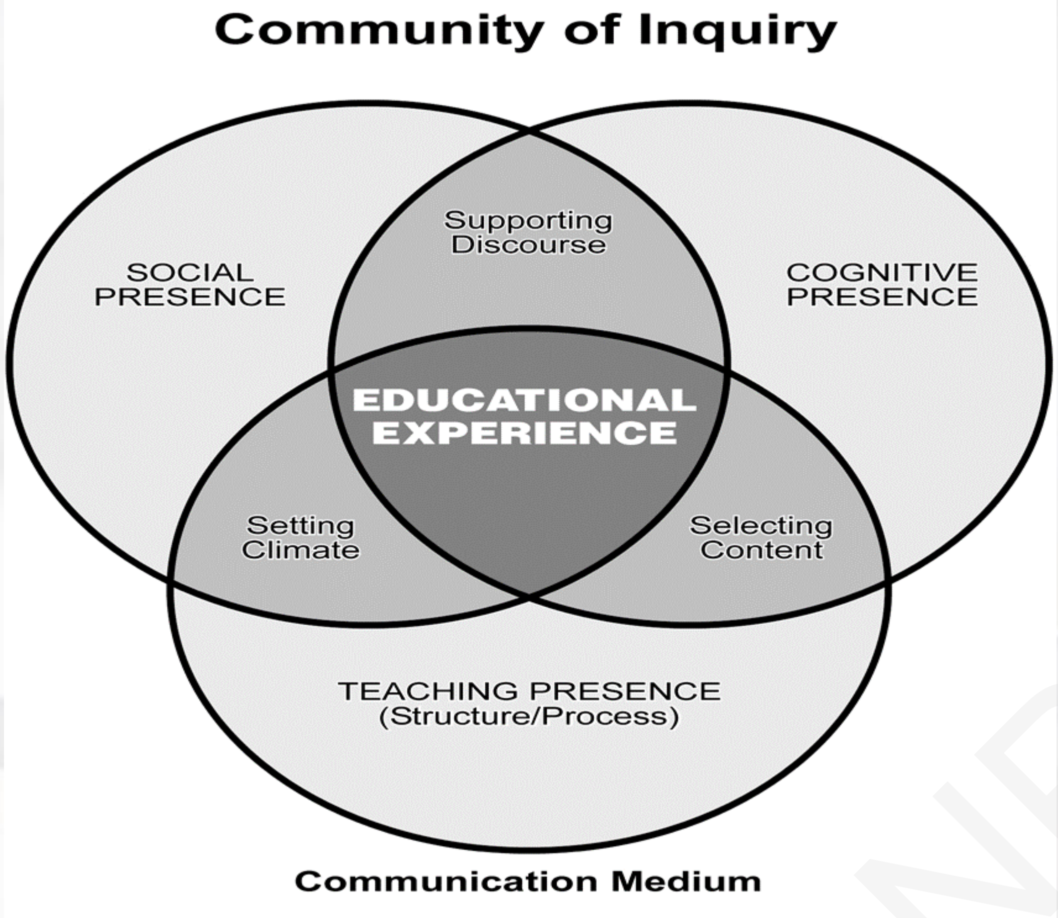
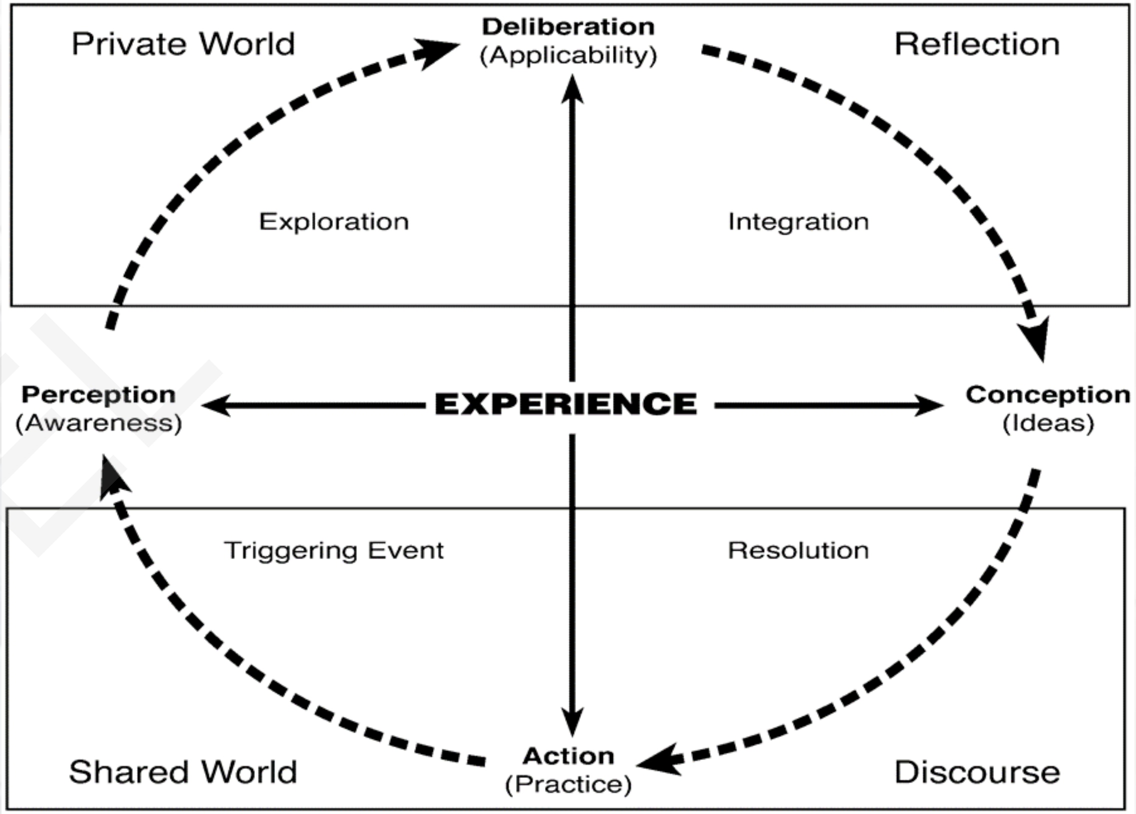


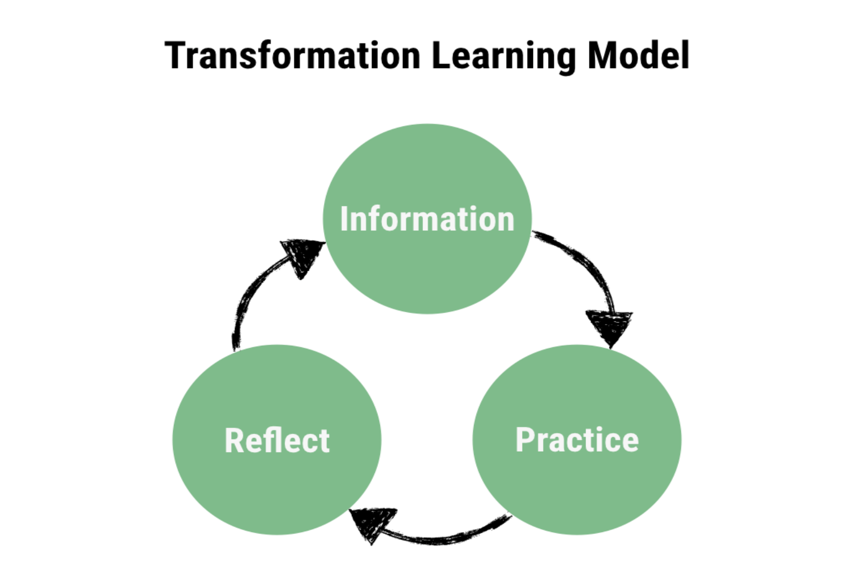
Figure 6 – Flipped Classroom Based on the Cognitive Flexibility Theory



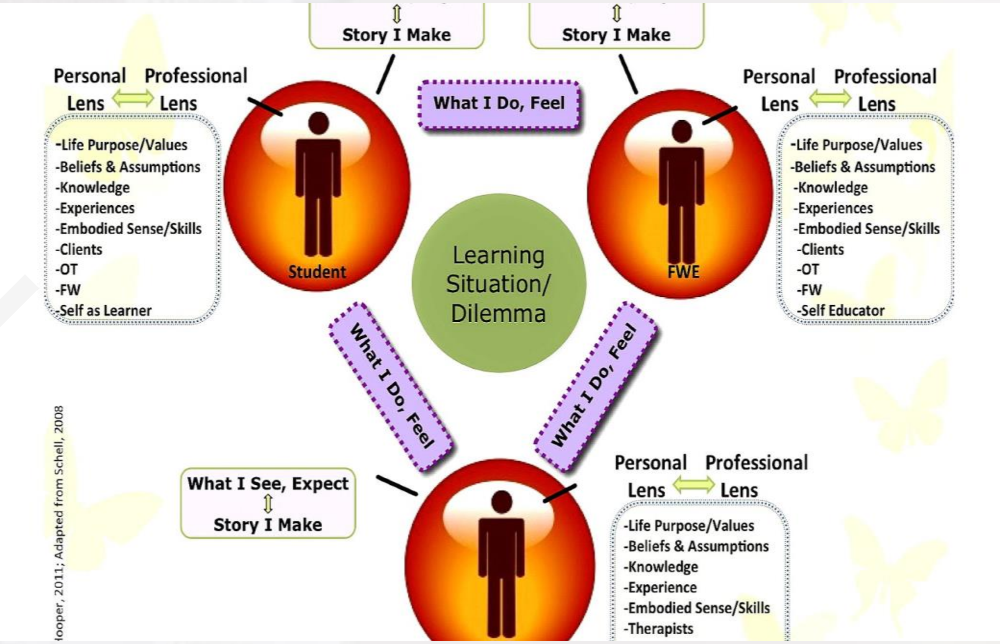


Community of Inquiry framework (CoI) research has provided the empirical evidence that the CoI framework represents a coherent set of articulated elements and respective models describing a higher learning experience applicable to wide range of learning environments (from face-to-face to online, from K-12 to higher education). It provides the means to understand and explore the relationships among the elements and learning, (Garrison, 2011).

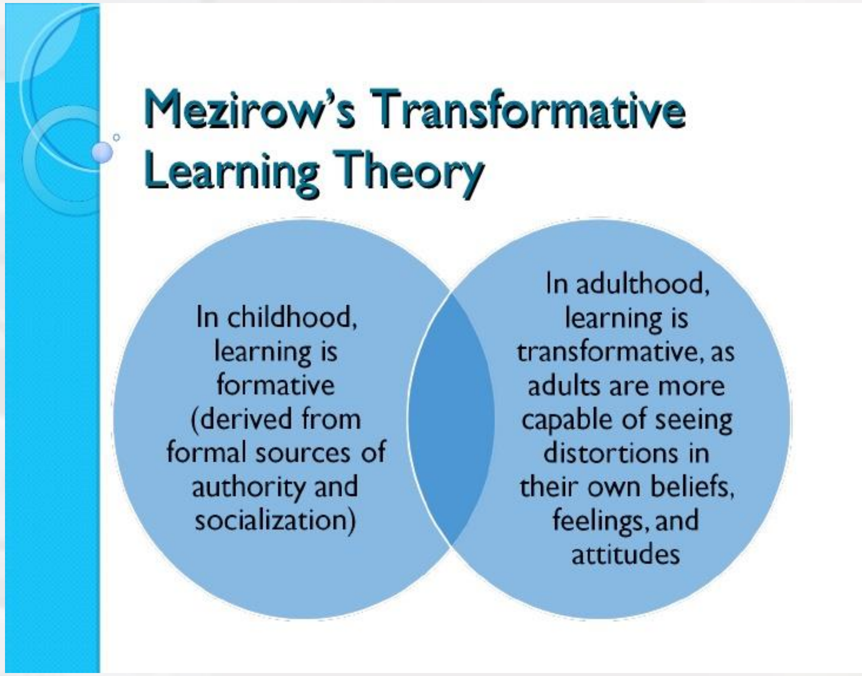
- **Transformative Learning-**
- Transformative learning is "the process of effecting change in a frame of reference" (Jack Mezirow, the father of transformative learning)
- A frame of reference includes a student's habit of the mind, as well as a personal point of view
- The habits of mind are affected by previous learning experiences and cultural norms, while the points of view are the student's personal beliefs and attitudes (Mezirow, 1997)
- **Mezirow** identified **four processes of learning**:
- 1. Elaborate an existing point of view
- 2. Establish new points of view
- 3. Transform previous point of view
- 4. Transform habits of the mind
- ✓ When a learner first engages with learning content or begins an assessment, he or she tends to look for evidence that supports his/her own beliefs and pre-conceived notions



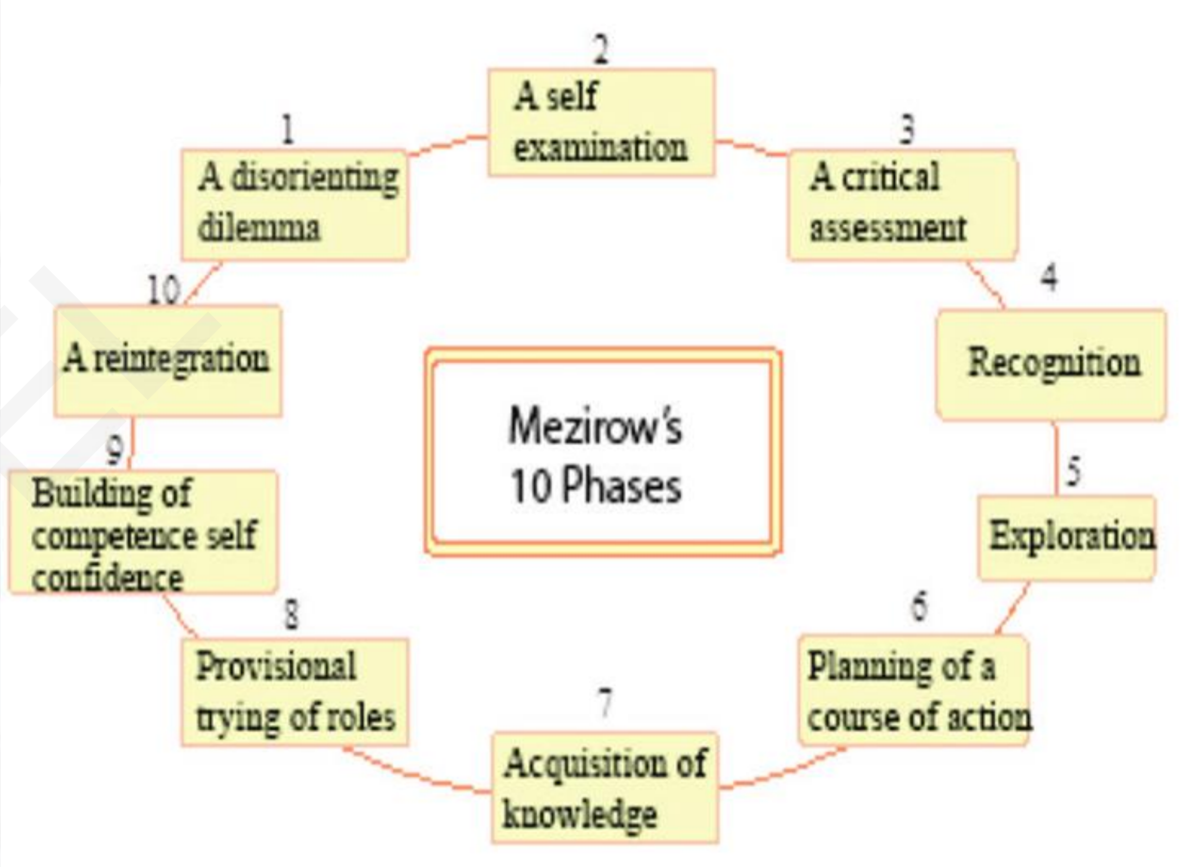
- The educational experience begins to transform the student and he starts to examine alternate points of view
- These alternate points of view may then replace or be added to the existing point of view to create a new point of view
- This transforms into a habit of the mind when the learner can learn to look at things differently
- This includes acknowledging potential biases of previous, as well as new points of view (Mezirow, 1997)
- When learning occurs the student interprets the new information based on previous experience, this best happens as a product of reflection on the learning itself.
- Reflection on learning includes making inferences, discriminating how the information meets or challenges pre-conceived notions, evaluating the information itself, and last, solving a problem or dilemma.



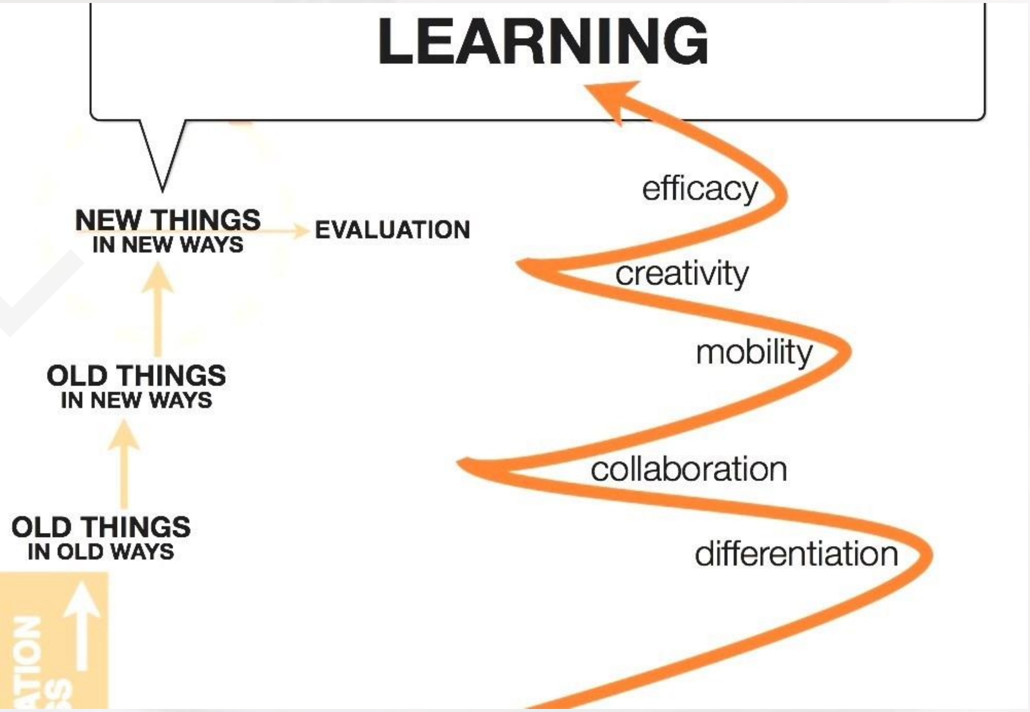
- This last stage can include deciding if the information meets the students' morals and ethical beliefs, or if they challenge these beliefs.
- This process is continual and throughout their lives, humans continue to evaluate both knowledge as actions.
- Discrimination between active interpretation of knowledge and reflective interpretation
- Active interpretation happens very quickly before all of the facts and nuances are evaluated
- Reflective interpretation of learning takes longer and is usually less biased because most or all aspects have been evaluated and filtered through the learner's experiences and beliefs;
- Sterling (2011) - Not everything a student learns spurs them to action, no matter how the faculty member has tried to include transformative education principles
- Certain concepts will speak to certain students, while some students will be able to relate to others.



- Mezirow(1978)- identified 10 steps that are required for transformative learning to occur These 10 steps are:
- a) a disorienting dilemma,
- b) self-examination,
- c) discontentment; realizing others are also discontent and have changed,
- d) evaluation of potential options,
- e) critical assessment of personal assumptions,
- f) experimenting with new roles,
- g) planning a course of action,
- h) attaining knowledge and skills to realize action plan,
- i) attainment of competence in new role,
- j) reintegration of new perspective

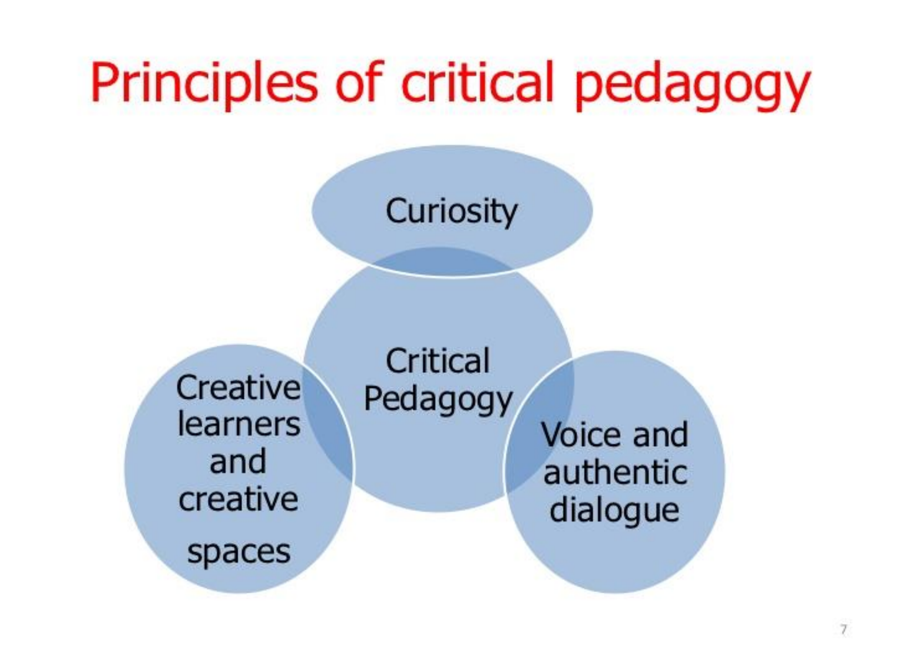


- Ison & Russell (2000)- identified two levels of change that are driven by learning; first and second order change
- First order change is the type of change that occurs with traditional pedagogies where lecture and testing are the primary modalities used
- In first order learning, students may learn the content for a test, but it never really makes any long-term impact on their lives
- What they have learned is quickly forgotten
- Second order change impacts both the way a student thinks and believes, as well as the way he acts
- This may be a service learning experience that assists a student in realizing what he wants for his career.
- Sterling called first order change as cognition and second order change meta-cognition



- Sterling added a third order change called epistemic learning; this form of learning changes a student’s worldview
- Mezirow (1990) - “Reflection is generally used as a synonym for higher order mental processes”
- It allows students to gain new understandings and appreciation
- He noted reflection is different from thinking because it requires additional analysis
- Reflective exercises are used as a key educational strategy in transformative learning to assist students in reaching second or even third order change.
- In transformative learning, assessments will include problem-based learning and reflective assignments that may take a great deal of student time and planning
- Peer-to-peer learning is an important part of authentic assessment since students will be required to collaborate in the real world on the job.

- Critical Pedagogy Theory
- Kucukaydin & Cranton (2012) - transformative learning is a theory in progress; proposed transformative learning is an extra-rational, post-modern epistemology where learners can critique each other's perspectives and knowledge with an open mind and effectively communicate differences
- Knowledge is subjective and through reflection of one's own knowledge truth is sought
- One's truth should be open to questioning that can include the creation of assignments that use deductive logic.
- Sterling (2011) recommends in order for transformative learning to occur, large classes of students must be broken down into small groups that can effectively interact
- The faculty member must create an environment conducive to online learning which may result in the faculty member being viewed less as an authority figure and more as a coach.

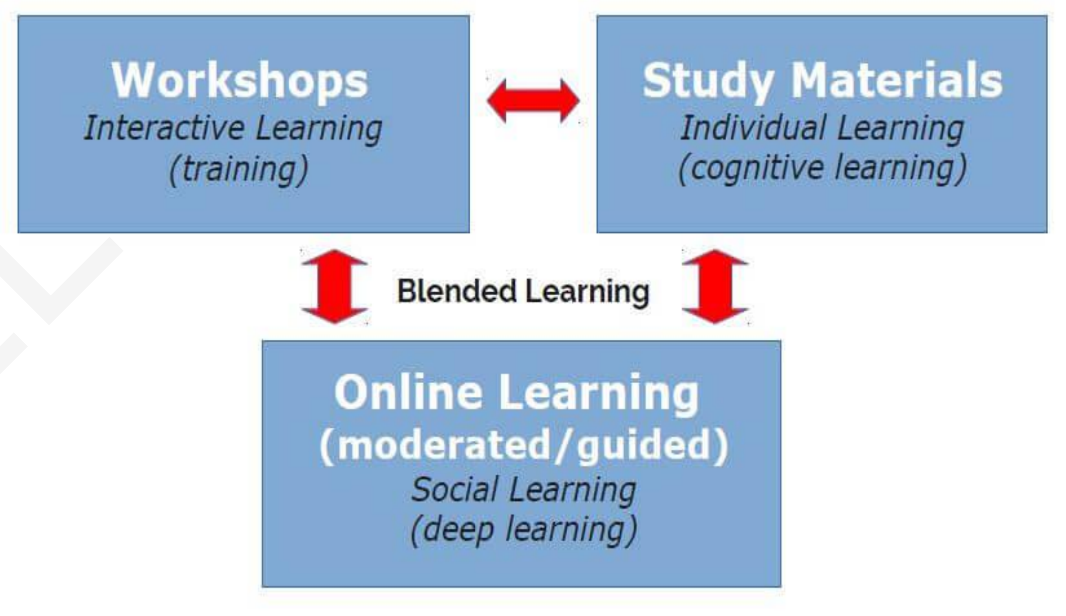


- Sterling - listed an intensive residential experience as an environmental factor that can enhance transformative learning
- Transformative learning can and does work in the asynchronous online environment as well
- Effective discourse can be done with synchronous meeting technology, as well as robust discussion forums that include Socratic questioning.
- Students must do-
- Tahiri (2010) -for transformative education to occur students must:
- a) acknowledge they are equal partners with the professor in the learning experience,
- b) be open for change,
- c) be willing to determine their own reality,
- d) be willing to share their life event,
- e) be willing to engage in critical reflection,
- f) show maturity on dealing with change

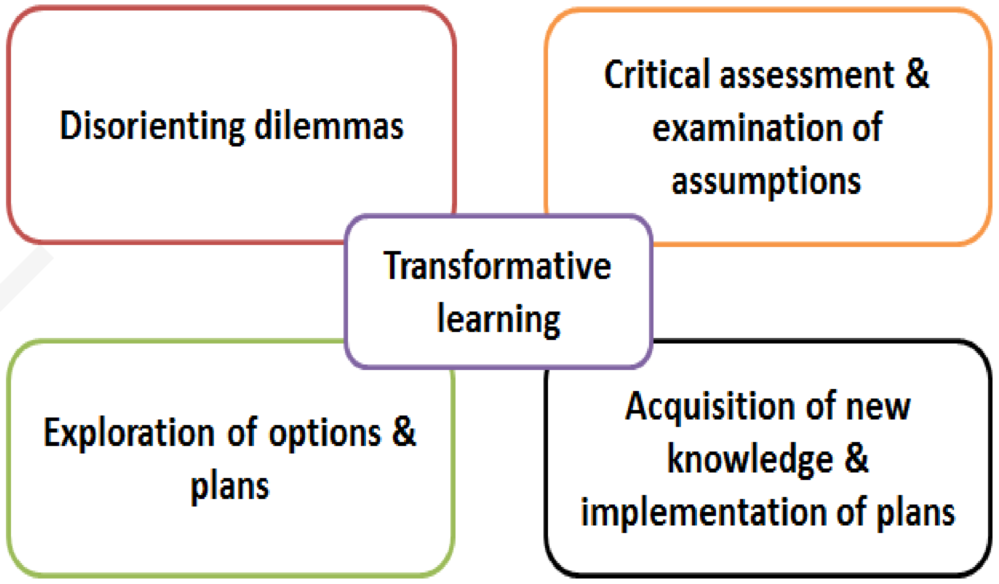
Key concepts of Critical Pedagogy

- Learning through discourse
- Critical Consciousness
- Reconceptualizing Literacy
- Honouring lived experience
- Historical context
- Praxis- both dialogue and action
- Reflection
- Direct honesty
- Identifying and actively countering hidden oppression
- Student Voice/ Empowerment
- A constructive rather than banking pedagogy of teaching
- Breaking down Eurocentric thought/ thinking
- Acknowledgement of the emotional nature
- Socialist vision / revolutionary agenda
- Hegemony

- The emancipatory philosophy has a central role in mainstream transformative learning theory, with the journey from one set of assumptions to another conceptualized in terms of liberation (Mezirow, 1991)
- . . . Any analysis of adult learning or adult learning gains must address both instrumental and communicative learning, including learning about oneself, as well as the nature, extent, and impact of critical reflection in both domains
- Magro(2009)-Blended approach—in Technical & Vocational Education-
- The focus to be on "technical" subject matter relating to occupational knowledge and personal development—both can "enhance and enrich" this kind of education
- Transformative learning theory is envisaged as a way to inform such development in TVET.



- As pedagogy, transformative learning provides a paradigm of education that fosters a powerful shift in beliefs or values
- As an outcome, it constitutes a new lens through which to see oneself or others
- Transformative learning can be both cognitive and imaginative; it can be collaborative and individually based projects
- It can include depth psychology alongside a more practical reflective approach
- Transformative learning is still an emerging theory
- Transformative learning begins when individuals reflect critically upon their assumptions of what they believe to be real, true, or right
- Critical reflection is the ongoing process of consciously or unconsciously reviewing and evaluating assumptions to clarify the meaning of experiences both individually and collectively.



- Nerstrom Transformative Learning Model (Figure 1-2014)
- The model is loosely based on Mezirow's (1978) phases of transformative learning.

Experience. Experience is everything that has occurred in our lifetime. It is the impetus of our learning and belief patterns. Experience stems from our environment and interactions with others, from which learning—such as knowledge, skills, attitudes, and insights—occurs.

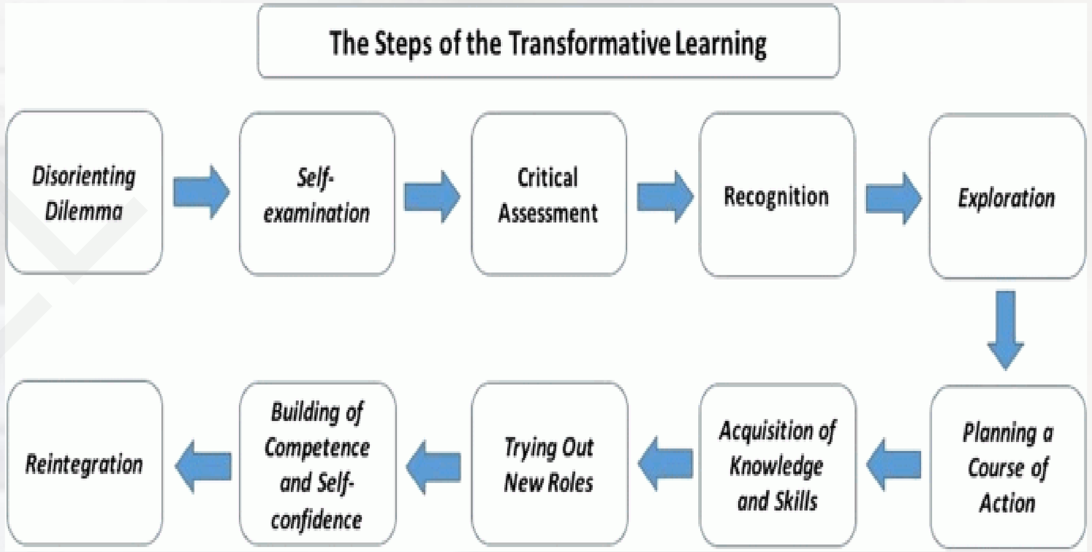
Assumptions. From experience we form, receive, and construct assumptions that become our values and beliefs. They are the lens through which we view the world.

Challenge Perspectives. New experiences, both cognitive and affective, combined with reflection, may lead us to challenge our deeply held assumptions and consider a new perspective.

Transformative Learning. Adopting and acting upon a new perspective, we view ourselves and others through a more encompassing lens. Transformative learning becomes a new experience leading to openness for it to occur again.

Figure 1. Nerstrom Transformative Learning Model

- Demonstrating the Model: Figure 1
- Five major themes emerged from this research:
- (a) examined prejudices—biases, stereotypes, and learned beliefs;
- (b) incidental experiences, with subthemes of increased self-confidence, renewed personal values, cultivated social involvement, and lasting friendships;
- (c)program structure fostering transformative learning, with subthemes of cohort and residential learning and traditional learning models;
- (d) reconceptualization of learning; and
- (e) transformed personhood (Nerstrom Transformative Learning)
- Are Students and faculty ready for transformative learning? ?????????



- **Contemplative Learning & Pedagogy-**
- Contemplative education is a philosophy of higher education that integrates introspection and experiential learning into academic study in order to support academic and social engagement, develop self-understanding as well as analytical and critical capacities, and cultivate skills for engaging constructively with others.
- The inclusion of contemplative and introspective practices in academia addresses an increasingly recognized imbalance in higher education:
- a lack of support for developing purpose and meaning, or for helping students "learn who they are, search for larger purpose for their lives, and leave college as better human beings“
- Generally, there are three main ways that contemplative practices are incorporated,
- firstly, in a remedial manner where say a simple breath exercise will be used to help students relax and orient their focus on class content and exercises.
- Secondly, the physiological, psychological, philosophical and religious foundations of the practices will be taught, and
- lastly, a contemplative orientation will be developed in the class room or across the entire institution.

- Contemplative Education is described as holistic and progressive and contrasted with mainstream or conventional forms of education that focus on the acquisition of knowledge, development of cognitive skills and individual achievement.

- The need for contemplative education is predicated on the failure of educational systems that accentuate a ‘curriculum of content’, over understandings of the whole student and teacher and the process of learning.

- **A pathway to flourishing and well-being**

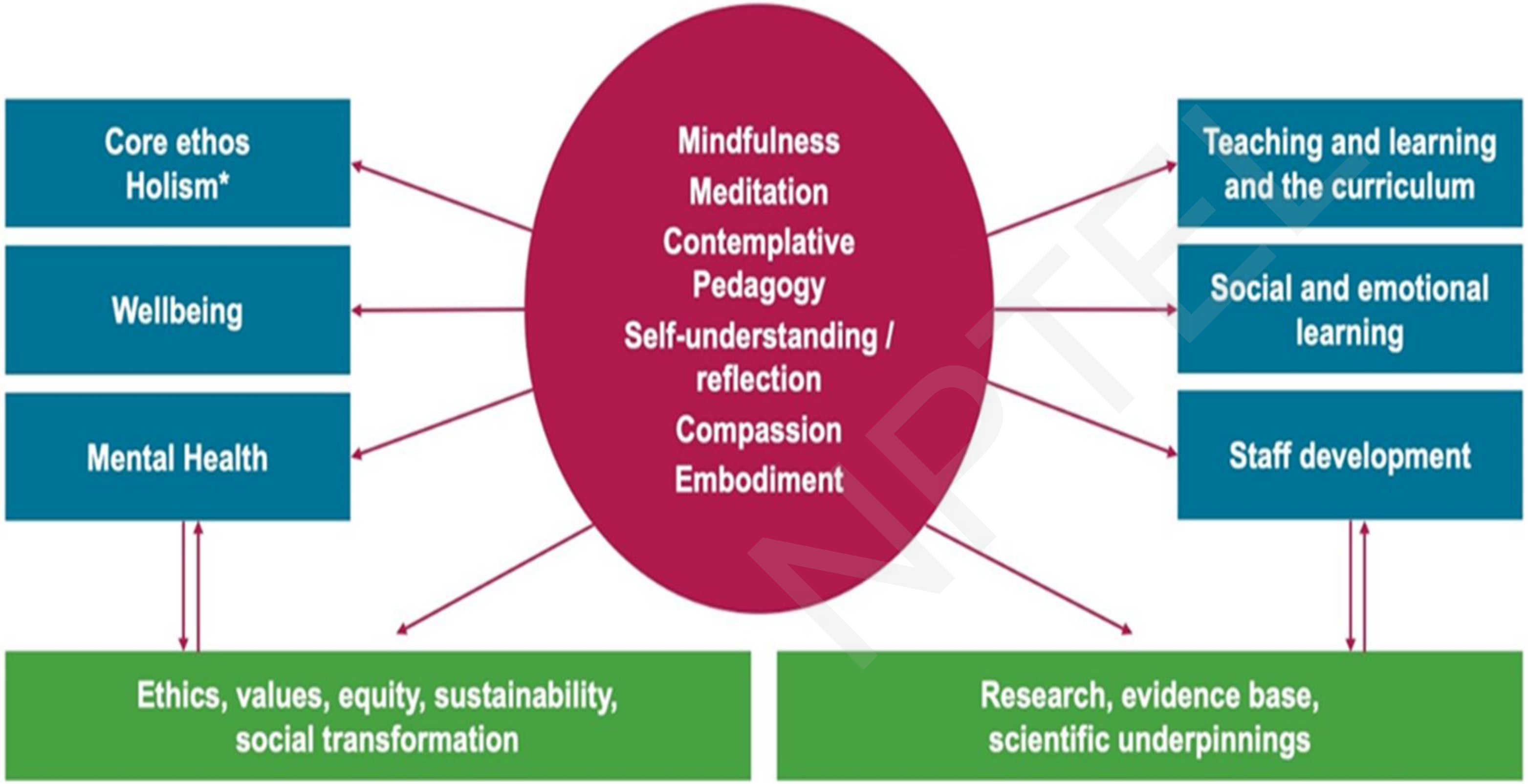
Many foreign language instructors have found that participating in the learning community has had a significant impact on their own dispositions toward students and teaching.

Contemplation in the arts and the sciences-

Contemplative practices and approaches are associated with numerous beneficial outcomes related to learning, such as increased emotional awareness and attentional control.

A WORKING MODEL OF CONTEMPLATIVE EDUCATION (CE)

- Core elements of CE
- Examples of the wider field in education to which MFN contributes
- Wider contribution to society and scientific foundations

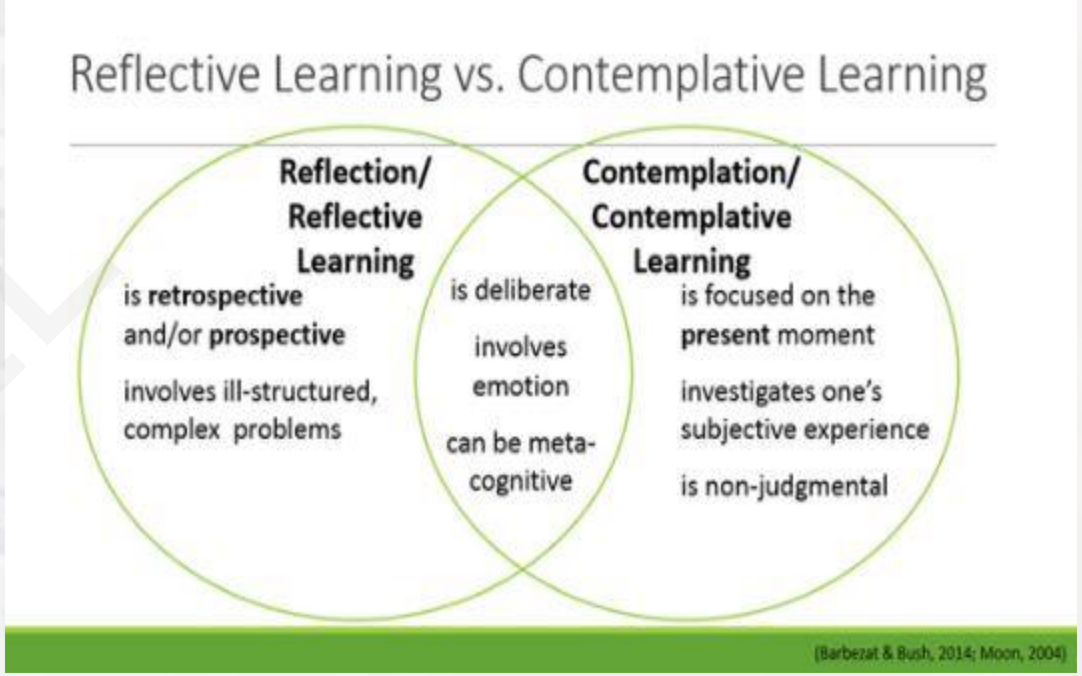


*„Holism" refers to joined up thinking – an approach that takes in the totality of the organisation as the focus of change efforts (not just the curriculum) and at all aspects of the person (social, emotional, physical, spiritual, cognitive) not just the cognitive.

Version: February 2020

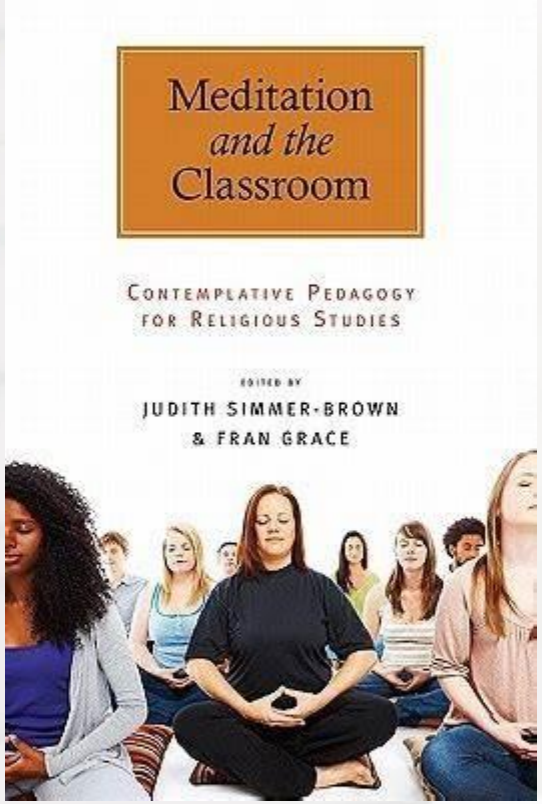
- Contemplative learning was education that created a new vision to life and humanity.
- It focused on cultivating realization in oneself, compassion and conscience to general people,
- Utilizing philosophy and religion to develop the mind and train oneself until one has conscience and intelligence can connect the various knowledge and apply them to be beneficial to oneself, others and society (Royal Institute 2008).
- Contemplative intelligence is a concept that would revive education to be education that develops learners in order to create equilibrium regarding knowledge and mind;
- Mindful and Contemplative Pedagogy is an approach to teaching and learning which encourages instructors and learners to be in the present moment, to fully engage in teaching and learning, and to achieve focus and attention in the classroom.

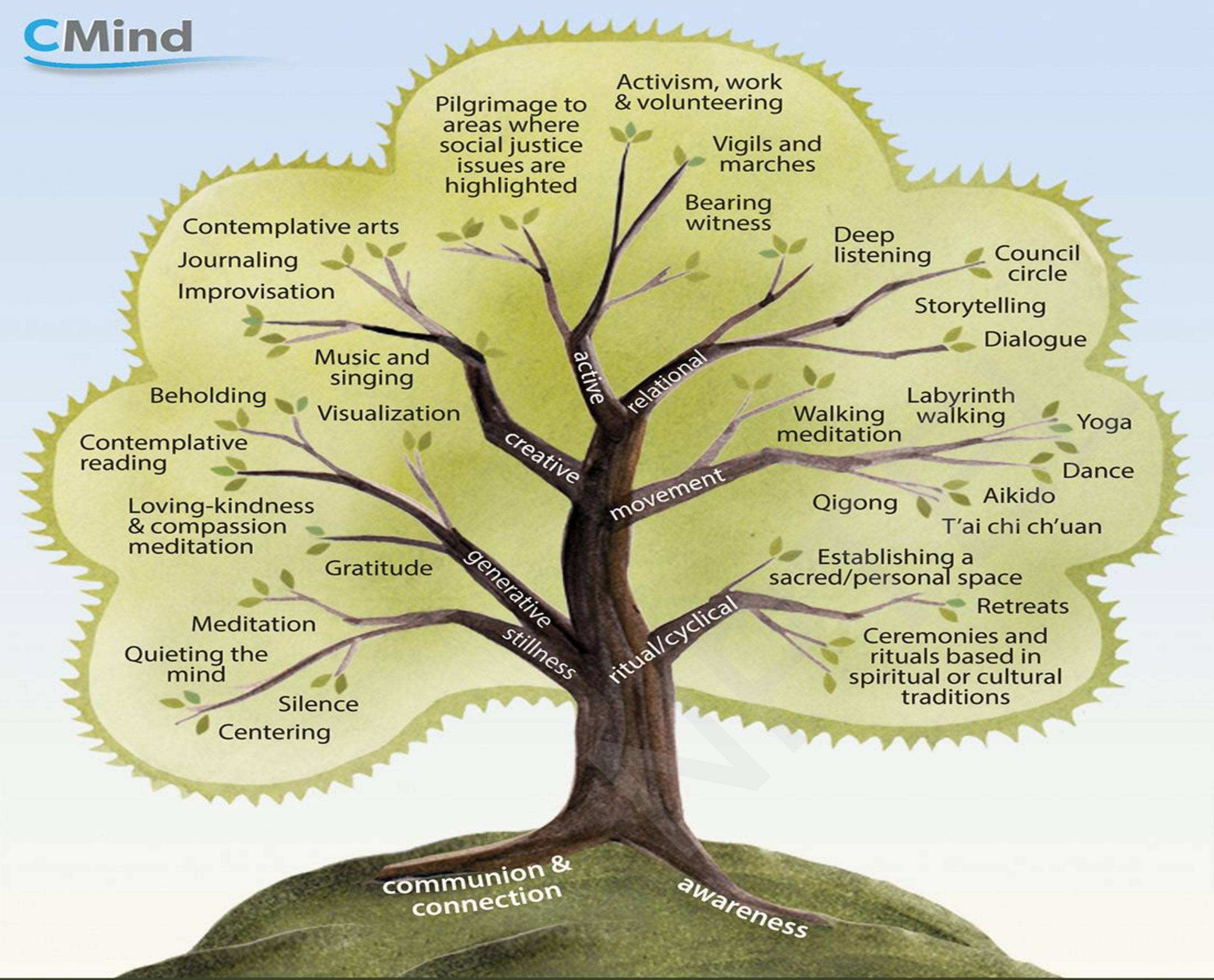
- As it should be in order to be a person with good virtue and to know oneself and to understand other people and can live together and to have lives that would create benefit with the society happily,
- Educational technology is another important part in managing contemplative learning;
- Educational technology is the knowledge in educational development and passing on knowledge from instructors to learners with the concept from A to Z.
- Contemplative pedagogy shifts the focus of teaching and learning to incorporate ‘first person’ approaches which connect students to their lived, embodied experience of their own learning.
- Students are encouraged to become more aware of their internal world and connect their learning to their own values and sense of meaning which in turn enables them to form richer deeper, relationships with their peers, their communities and the world around them.



- The forms that contemplative practice can take within the classroom are numerous, including: the use of meditation or mindfulness techniques which improve concentration and allow deeper contemplation of the material being taught; writing techniques such as free writing and journaling and reading exercises;
- Deep listening activities which encourage students to find a voice for their own thoughts and connect more deeply with their peers; and the use of ceremony and ritual to incorporate an element of the sacred into teaching and encourage fresh perspectives.
- It incorporates ways of teaching and learning that can provide a very different learning experience by opening up new ways of knowing.
- This is achieved by moving beyond a technical, scientific training to incorporate body, mind and spirit by allowing the space for students to incorporate who they are and to understand how they are changed by what they learn.

- Contemplative practices can help students tap into their emotional reaction to course materials and to confront difficult course topics, allowing them to engage more openly and holistically with the course material and their peers in the class.
- Contemplative methods expand the traditional focus of teaching and learning to include:
- Focus and attention building: Learning is enhanced as students are attentive to what and how they are learning.
- Deeper understanding of and connection to course materials: Learning becomes more meaningful and relevant as students reflect upon course material and make connections to their experiences.
- Compassion and connection to others: Learners develop empathy and interpersonal connection.
- Self-inquiry, personal meaning, and creativity: Learners engage in a process of self-inquiry to deepen their relationship to their learning, tap into their creativity and insight, and develop more awareness about their own learning processes.
- Instructors can build in opportunities for students to develop deeper understandings of course material by giving them time to reflect on what they are learning, how they are learning (i.e., their learning process), and by making connections between their own experiences and course material.





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Thank You!!